

# Real Origins of Financial Structure: The Role of Household and Firm Heterogeneity <sup>\*</sup>

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## Abstract

Financial systems channel funds from savers to borrowers through a mix of bank- and market-based intermediation. We show that financial-sector structure is shaped, in large part, by real-sector characteristics that generate demand for heterogeneous savings and borrowing instruments. Across countries, economies with younger, wealthier, and more unequal household sectors rely less on banks, as do economies whose firm sectors feature wider size distributions, more intangible-intensive production, and larger service sectors. Using U.S. microdata, we show that these patterns reflect financial decisions that vary systematically across households and firms. Motivated by this evidence, we develop and estimate a model with heterogeneous households and firms in which different types of intermediaries compete to satisfy savings and borrowing demand. We estimate how household and firm demand varies with real-sector characteristics using U.S. microdata, and calibrate remaining parameters to match asset prices and aggregate balance sheets. Counterfactuals show that rising household wealth, aging, and wealth concentration since the 1980s reduce the role of bank deposits, while firm heterogeneity primarily affects the composition of credit demand. These forces provide a demand-side foundation for the growth of alternative financing: a wealthier household sector demands more market-based claims, while changes in firm composition shift relative demand from bank toward nonbank finance, including forms of private credit and private equity. Cross-country real-sector counterfactuals explain a meaningful component of international variation in bank funding and credit shares, while leaving substantial residual dispersion, especially on the credit side.

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Financial systems perform a common task—channeling funds from savers to borrowers—but they do so through heterogeneous institutional structures. Some economies are more bank-centered, others rely more heavily on market-based intermediation, and even within a given economy the boundary between banks and markets shifts over time. Standard explanations emphasize regulation, financial development, and frictions within the intermediary sector. But those forces operate on a more basic object: the heterogeneous households and firms the financial system serves. Households and firms differ not only in how much they save or borrow, but also in which financial claims they use and how easily they substitute across them. As a result, the structure of intermediation is not simply a feature of the financial sector itself; it is an equilibrium outcome shaped by the characteristics of savers and borrowers.

In this paper, we show that the boundary between banks and markets is shaped in large part by the real sector. Banks and markets do not intermediate generic savers and borrowers: they serve households and firms whose demand for financial claims varies systematically with wealth, age, size, and industry. Household characteristics determine whether savings are held as deposits or market-based claims, and whether borrowing is obtained from banks or nonbank lenders. Firm characteristics determine the demand for liquid assets and the choice between bank credit and market-based debt. These demand forces jointly determine banks' funding base, credit role, and balance-sheet composition. This provides a demand-side foundation for understanding the growth of alternative financing. A wealthier household sector supplies more capital to market-based claims rather than deposits, while changes in firm composition shift the relative demand for bank versus nonbank credit. These forces create the demand conditions under which private credit, private equity, and other nonbank intermediaries can expand alongside, and in some cases substitute for, traditional deposit-funded banks.

We motivate this perspective with cross-country evidence. Across a broad sample of economies, financial structure varies systematically with the composition of the real sector. Economies with younger, lower-net-worth household sectors and smaller firms are more bank-centered, while economies with wealthier households, older populations, larger firms, and more service- and intangible-intensive production rely more heavily on market-based finance. These patterns survive controls for legal origin, financial development, and standard institutional measures, and are visible in both levels and changes over time. The macro evidence is suggestive, but it is not sufficient for quantification. It cannot distinguish demand-side substitution from intermediary-side institutions, nor can it identify the relevant

elasticities. To make progress, we build a structural model disciplined by U.S. microdata, and then use the model to ask how changes in the real sector reshape equilibrium intermediation.

The model is organized around three margins: who saves, who borrows, and through whom funds are intermediated. Households are indexed by net worth, age, and income. They allocate saving across deposits, money market fund shares, bonds, and equity, and allocate borrowing across bank and nonbank lenders subject to a cash-flow constraint. Firms are indexed by industry and size. They allocate assets between real capital and financial assets, split financial assets between deposits and money market funds, and finance themselves with a composite of bank and nonbank debt. The intermediary sector consists of banks, money market funds, and nonbank lenders. Banks issue deposits and make loans under monopolistic competition, subject to liquidity and capital wedges; money market funds and nonbank lenders are competitive institutions that transform marketable securities into MMF shares and nonbank credit. Government and rest-of-world sectors close the model by supplying public safe assets and absorbing marketable claims. The key empirical object is how the preference weights in these demand systems vary with observable household and firm characteristics.

Two sets of U.S. microdata facts discipline this heterogeneity. On the household side, the Survey of Consumer Finances reveals that saving and borrowing exhibit sharply different wealth gradients. Deposit shares are high among low-net-worth households and fall steeply with wealth as richer households substitute toward fixed-income and equity claims. By contrast, the bank share of borrowing rises with wealth: low-net-worth households save through banks, while high-net-worth households are more likely to borrow from them. Age also matters for both portfolio composition and lender choice. These patterns imply that household saving and borrowing must be modeled separately, and that observable household characteristics shift demand across financial instruments.

On the firm side, Census Quarterly Financial Report data show that firm balance sheets vary systematically with size and industry. Smaller firms hold larger financial buffers and rely more heavily on deposits within those buffers, while larger firms substitute toward market-based liquidity. On the liability side, bank reliance is non-monotone in firm size: mid-sized firms have the highest bank-loan shares, while the largest firms rely more heavily on bond markets and other nonbank credit. These facts motivate a firm block in which size and industry affect the allocation between real and financial assets, the composition of liquid assets, leverage, and the allocation of debt across bank and nonbank credit.

The estimation separates elasticities from heterogeneity. Across-class elasticities are iden-

tified from time-series variation in relative returns using aggregate Financial Accounts data and policy-related instruments. Given these elasticities, U.S. microdata discipline household and firm demand for savings and borrowing instruments. SCF data identify how household net worth, age, and other characteristics shift demand across deposits, market-based saving instruments, bank borrowing, and nonbank borrowing. QFR data identify how firm size and industry shift demand across liquid assets, bank credit, and market-based debt. The remaining model parameters are calibrated to match equilibrium prices and aggregate balance-sheet quantities. This approach uses U.S. microdata to discipline the real-sector demand system, rather than calibrating the model to match cross-country outcomes.

We use the estimated equilibrium model to quantify how real-sector heterogeneity shapes financial intermediation. In all counterfactuals, intermediary technology, regulatory wedges, preference elasticities, policy variables, and institutional features are held fixed at their 2019 U.S. estimates. We vary only the distribution of household and firm types. The exercises therefore ask how financial structure changes when the real sector changes, holding the financial environment fixed.

The first counterfactual studies comparative statics around the 2019 U.S. economy. Household characteristics primarily affect banks through the funding side and household credit demand. Higher household wealth reduces deposit reliance because richer households allocate more savings directly to market-based claims; at the same time, it raises borrowing capacity and credit demand, shifting bank balance sheets from securities toward loans. Wealth concentration and aging operate mainly through portfolio composition. Firm characteristics operate through a different margin: the firm-size distribution changes the demand for bank versus nonbank credit and therefore affects the asset side of banks' balance sheets.

The second counterfactual studies how real-sector change affected U.S. financial structure between 1983 and 2019. The long-run shift is primarily a household-side phenomenon in the model. Rising wealth, aging, and wealth concentration increase demand for directly held market claims, reduce the share of saving absorbed by bank deposits, and allow banks to allocate more of their balance sheets to loans rather than securities. Firm heterogeneity primarily affects the composition of credit demand, especially through the firm-size distribution. Together, these forces provide a demand-side foundation for the growth of alternative financing: a wealthier household sector demands more market-based claims, while changes in firm composition shift the relative demand for bank versus nonbank finance, including forms of private credit and private equity.

The third counterfactual uses cross-country real-sector characteristics to construct coun-

terfactual U.S. economies. For each country, we shift the U.S. household and firm distributions toward that country’s real-sector characteristics while keeping U.S. intermediary institutions fixed. This is not a full cross-country calibration; it asks how much cross-country variation in financial structure can be generated by real-sector demand alone. The answer is asymmetric across margins. On the funding side, real-sector characteristics explain a meaningful component of international variation in deposit reliance. On the credit side, real-sector characteristics move the model in the right direction but leave substantial residual dispersion. This residual variation is consistent with an important role for intermediary-side institutions and financial-market development.

Taken together, the counterfactuals show that real-sector characteristics are an important state variable for financial structure. They shape banks’ funding base, the relative demand for bank versus nonbank credit, and banks’ own balance-sheet composition. At the same time, real-sector demand is not a substitute for institutional explanations. Within the United States, changes in household and firm characteristics account for meaningful changes in intermediation and help rationalize the demand-side conditions under which alternative financing expands. Across countries, real-sector characteristics explain a structural component of bank funding and credit shares, while institutional differences and financial-market development account for additional dispersion, especially on the credit side.

Our paper contributes to three related literatures. First, we contribute to work on the structure of financial intermediation by showing that the boundary between banks and markets is partly an equilibrium response to real-sector demand. Second, we contribute to the literature on household and firm financial choice by showing how micro-level portfolio and borrowing heterogeneity aggregates into macro-level financial structure. Third, we contribute to debates on financial development, nonbank intermediation, and alternative financing by emphasizing the demand-side conditions under which market-based and nonbank finance become more important. More broadly, the paper offers a framework for studying how changes in the real economy reshape not only portfolios and credit choices, but the organization of the financial system that intermediates between savers and borrowers.

## 1 Cross-Country Macro Evidence

We begin by documenting that financial-sector structure varies systematically with real-sector characteristics across countries. Two patterns motivate the model. First, the share of household financial assets held as deposits varies sharply across countries with the wealth

distribution and the age structure of the household sector. Second, the share of credit intermediated through banks varies with the industrial composition of the firm sector. Both patterns survive standard panel controls. Our point is not that these macro correlations identify any single mechanism: they reflect a mixture of household preferences, firm production technology, intermediary institutions, regulation, and tax policy. Rather, the patterns establish that the same real-sector characteristics that drive cross-sectional financial decisions in microdata also covary with aggregate financial structure across economies, supporting the broader argument that the boundary between bank-based and market-based finance is an equilibrium response to real-sector demand.

### 1.1 Saving side: deposit shares vs. household wealth and age

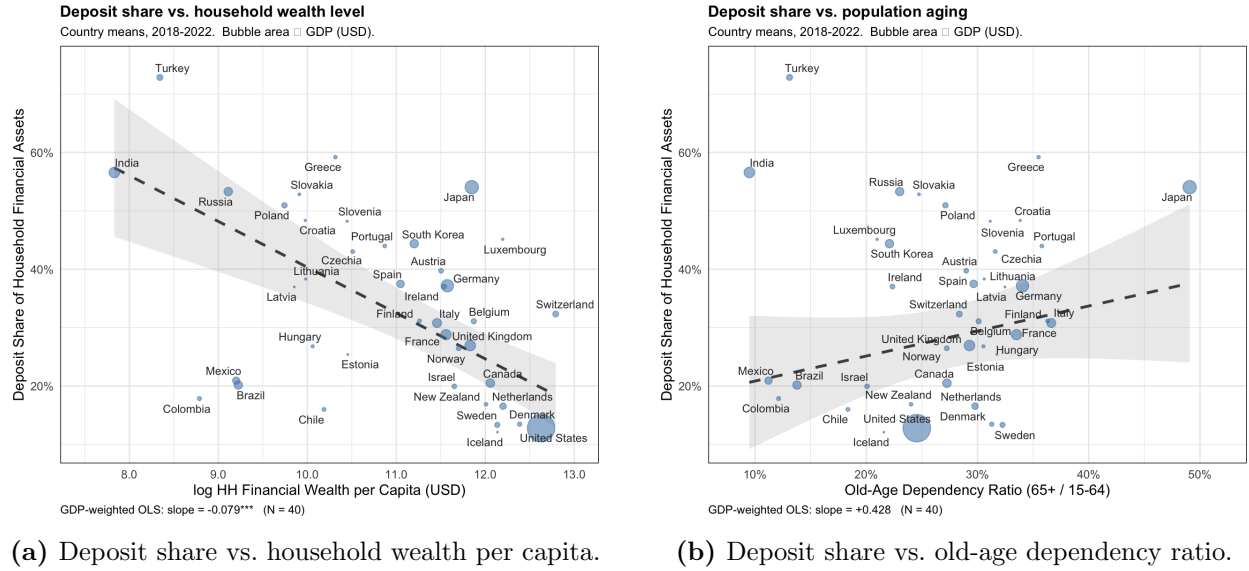
Figure 1 plots the cross-country relationship between the deposit share of total household financial assets and two real-sector characteristics: log household wealth per capita (Panel a) and the old-age dependency ratio (Panel b). The wealth gradient is economically large and negative: a one-standard-deviation increase in household wealth per capita is associated with a roughly 6 percentage-point reduction in the deposit share. The dependency-ratio gradient is positive and of similar magnitude. Wealthier and demographically younger societies hold smaller shares of their household financial assets in deposits.

### 1.2 Borrowing side: bank-loan shares vs. firm composition

Figure 2 plots the cross-country relationship between the bank-loan share of total firm credit and two firm-side characteristics: the services share of gross value added (Panel a) and intangible-investment intensity (Panel b). Both gradients are economically large and negative. Countries with services shares roughly 10 percentage points higher than the OECD median have bank-loan shares roughly 5–7 percentage points lower, and a gradient of comparable magnitude appears for intangible-intensive economies. Economies whose firm sectors generate more service-based and intangible-based value added rely less on bank intermediation, even before conditioning on year fixed effects or other controls.

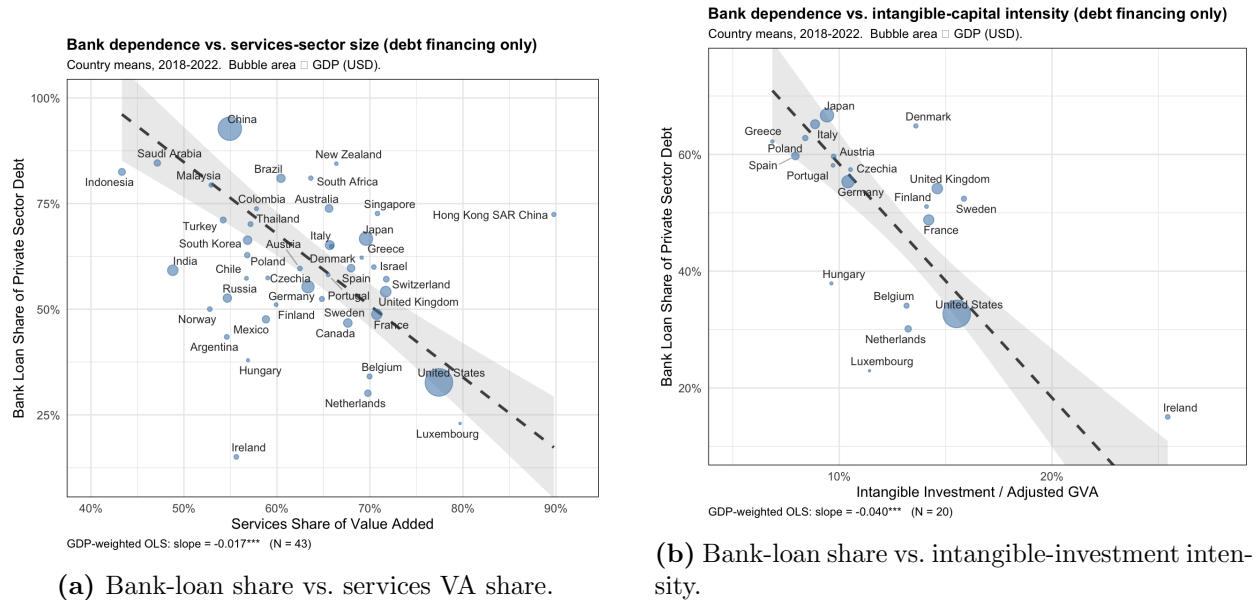
### 1.3 Panel regressions

The cross-country gradients survive year fixed effects, clustered standard errors, and standard institutional controls. Table 1 reports panel regressions of the deposit share (columns



**Figure 1: Deposit share of household financial assets and household real-sector characteristics.** Each panel plots, for a cross-section of forty countries averaged over 2018–2022, the deposit share of total household financial assets against a real-sector characteristic: log household wealth per capita in Panel (a), and the old-age dependency ratio in Panel (b). Each bubble represents a country and is sized in proportion to nominal GDP; the dashed line is a GDP-weighted ordinary least squares fit, with the shaded band the corresponding 95% confidence interval. The deposit share is constructed as the ratio of household holdings of currency and deposits to total household financial assets, both measured at market value on the asset side of the household balance sheet. Where available, the underlying balance-sheet data are taken from Eurostat’s annual financial accounts of households and non-profit institutions serving households (`nasa_10_f_bs`, sectors S14 and S15); for countries outside the Eurostat coverage we use the OECD’s annual financial balance sheets (Table 720R), preferring the consolidated household sector (S1M) and falling back to households alone (S14) where S1M is unavailable. Household wealth per capita is computed as total household financial assets in U.S. dollars (same OECD source) divided by total population from the World Bank’s World Development Indicators. The old-age dependency ratio is the ratio of the population aged 65 and over to the population aged 15–64, computed from the United Nations World Population Prospects (2024 revision, medium variant). Country GDP for bubble sizing is taken from the World Bank’s World Development Indicators.

1–2) and the bank-loan share (columns 3–4) on the four real-sector covariates, with and without two sets of standard institutional controls: La Porta et al. (LLSV) legal-origin dummies and the World Governance Indicators (rule of law, regulatory quality, government effectiveness). Two patterns stand out. First, household-side covariates—log household wealth per capita and the old-age dependency ratio—are the dominant predictors of the deposit share, with quantitatively large and statistically significant coefficients that strengthen rather than vanish under the institutional controls (column 2). Second, on the bank-loan share the household-wealth coefficient is marginal in the baseline specification but becomes significantly negative once the controls are introduced; intangible-investment intensity remains significantly negative throughout (column 4). The pattern echoes what we will see in the



**Figure 2: Bank-loan share of total firm credit and firm-sector real characteristics.** Each panel plots the bank-loan share of total credit to the private non-financial sector against a firm-side characteristic: the services share of gross value added in Panel (a), and intangible investment as a share of (intangible-adjusted) gross value added in Panel (b). Country means are computed over 2018–2022, with samples of forty-three countries (Panel a) and twenty countries (Panel b); the smaller sample in Panel (b) reflects the more limited country coverage of the intangible-investment series. Each bubble represents a country and is sized in proportion to nominal GDP; the dashed line is a GDP-weighted ordinary least squares fit, with the shaded band the corresponding 95% confidence interval. The bank-loan share is constructed from the Bank for International Settlements’ Total Credit Statistics (*WS\_TC*), as the ratio of bank loans to the private non-financial sector to total credit to the private non-financial sector from all lenders. Both numerator and denominator are quarterly end-of-period stocks at market value, adjusted for series breaks, and reported as percent of GDP; we average over the four quarters of each year. The services share of gross value added is taken from the World Bank’s World Development Indicators (series *NV.SRV.TOTL.ZS*), defined to include the ISIC divisions 50–99 (wholesale and retail trade, transport, accommodation, finance, real estate, and other services). Intangible investment is the sum of the Corrado, Hulten, and Sichel intangible asset categories—research and development, software and databases, branding, design, and organizational capital—scaled by adjusted gross value added, defined as gross value added with newly capitalized intangibles added back to ensure consistency with the numerator. The intangible series are taken from Global INTAN-Invest (sponsored by the World Intellectual Property Organization), which extends the Corrado–Hulten–Sichel methodology to a wide cross-section of countries. Country GDP for bubble sizing is taken from the World Bank’s World Development Indicators.

model below: the funding side of intermediation is shaped primarily by households, while the borrowing side is shaped primarily by firms.

**Bridge to the model.** The macro evidence establishes that financial structure covaries systematically with real-sector characteristics in the directions predicted by a heterogeneous-agent view of intermediation. But cross-country regressions cannot identify the underlying

**Table 1: Real-sector characteristics and bank reliance, with institutional controls.** The table reports panel regressions of the deposit share of household financial assets (columns 1–2) and the bank-loan share of total credit to the private non-financial sector (columns 3–4) on real-sector covariates, in a country-year panel covering 1995–2022. Columns (2) and (4) additionally include the LLSV legal-origin dummies (with English as the omitted reference) and the three core World Governance Indicators (rule of law, regulatory quality, government effectiveness), reported in the lower panel. Each column includes year fixed effects; standard errors, in parentheses, are clustered by country. Asterisks \*, \*\*, \*\*\* denote significance at the 10%, 5%, and 1% levels. The deposit share is constructed from Eurostat’s annual financial accounts of households (`nasa_10_f_bs`) and the OECD’s annual financial balance sheets (Table 720R). The bank-loan share is from the Bank for International Settlements’ Total Credit Statistics (`WS_TC`). Log household wealth per capita is constructed from the OECD financial balance sheets and World Bank WDI (population). The old-age dependency ratio is from the UN World Population Prospects (2024 revision). The services value-added share is from World Bank WDI (`NV.SRV.TOTL.ZS`). Intangible investment as a share of adjusted gross value added is from Global INTAN-Invest. Legal origin follows La Porta, López-de-Silanes, Shleifer, Vishny (LLSV); WGI series are from the World Bank’s Worldwide Governance Indicators (2023 update).

|                        | Deposit share        |                      | Bank loan share      |                     |
|------------------------|----------------------|----------------------|----------------------|---------------------|
|                        | (1)                  | (2)                  | (3)                  | (4)                 |
|                        | Baseline             | + Controls           | Baseline             | + Controls          |
| log HH wealth / capita | -0.117***<br>(0.031) | -0.170***<br>(0.027) | -0.055<br>(0.041)    | -0.125**<br>(0.044) |
| Top-1% wealth share    | 0.093<br>(0.144)     | -0.587***<br>(0.195) | -0.296<br>(0.477)    | -0.551<br>(0.390)   |
| Old-age dep. ratio     | 1.23***<br>(0.213)   | 0.530**<br>(0.190)   | 0.838***<br>(0.240)  | 0.872**<br>(0.343)  |
| Services VA share      | 0.001<br>(0.004)     | 0.007<br>(0.006)     | 0.0004<br>(0.003)    | 0.008<br>(0.005)    |
| Intangible inv./GVA    | -0.004<br>(0.007)    | -0.000<br>(0.003)    | -0.016***<br>(0.005) | -0.013**<br>(0.005) |
| Legal origin (LLSV)    | —                    | Yes                  | —                    | Yes                 |
| WGI (RL, RQ, GE)       | —                    | Yes                  | —                    | Yes                 |
| Year FE                | Yes                  | Yes                  | Yes                  | Yes                 |
| Observations           | 333                  | 333                  | 255                  | 255                 |

micro elasticities, separate composition from preferences, or speak to counterfactuals. To make those identifications, we now turn to a structural model that embeds heterogeneous households and firms, lets them face heterogeneous intermediaries, and is disciplined by US microdata. We document the relevant cross-sectional facts at the start of each sector subsection, where they directly motivate the modeling choices.

## 2 Model Overview and Flow of Funds

We now turn to a structural model of how funds flow from savers to borrowers through a heterogeneous intermediary sector. The model is static, with a “morning” and “evening” period, and consists of: a household sector and a firm sector that save and borrow; three types of financial intermediaries—banks, money market funds (MMFs), and non-bank lenders (NBLs)—that compete to satisfy that demand; and a government and rest-of-world sector that close the model. This section gives a verbal description of the sectors and the financial instruments that link them. The decision problems are developed in the sections that follow.

**Households.** Heterogeneous households are indexed by a type  $i$  drawn from a distribution  $F^h(\cdot)$ . A type encodes whatever observable characteristics shift household preferences over saving and borrowing products. They derive utility over consumption and the returns of differentiated savings products. This captures a reduced-form consumption–savings decision where different classes of financial assets—deposits, MMF shares, bonds, and firm equities—satisfy different needs such as safe transaction services or risky returns. Households finance consumption and savings with their net worth and borrowing in differentiated lending products—bank and non-bank loans—subject to a cash-flow constraint.

**Firms.** Heterogeneous firms are indexed by a type  $i$  drawn from a distribution  $F^f(\cdot)$ . A type encodes whatever observable characteristics shift firm preferences over real and financial assets and over bank versus non-bank debt. Firms invest their assets in a combination of productive real assets and financial assets—deposits and MMF shares—which together they use to produce cashflows. Firms are financed with debt and equity. Their debt financing is a composite of bank and non-bank loans, which are imperfect substitutes, representing the idea that direct bank lending and market-based lending are potentially suited toward the financing of different assets due to differences in screening, monitoring, and enforcement. Firms have a reduced-form optimal equity-to-assets ratio from which they may deviate in order to take advantage of cheap debt financing. Firm scale is determined by an upward-sloping availability of real assets.

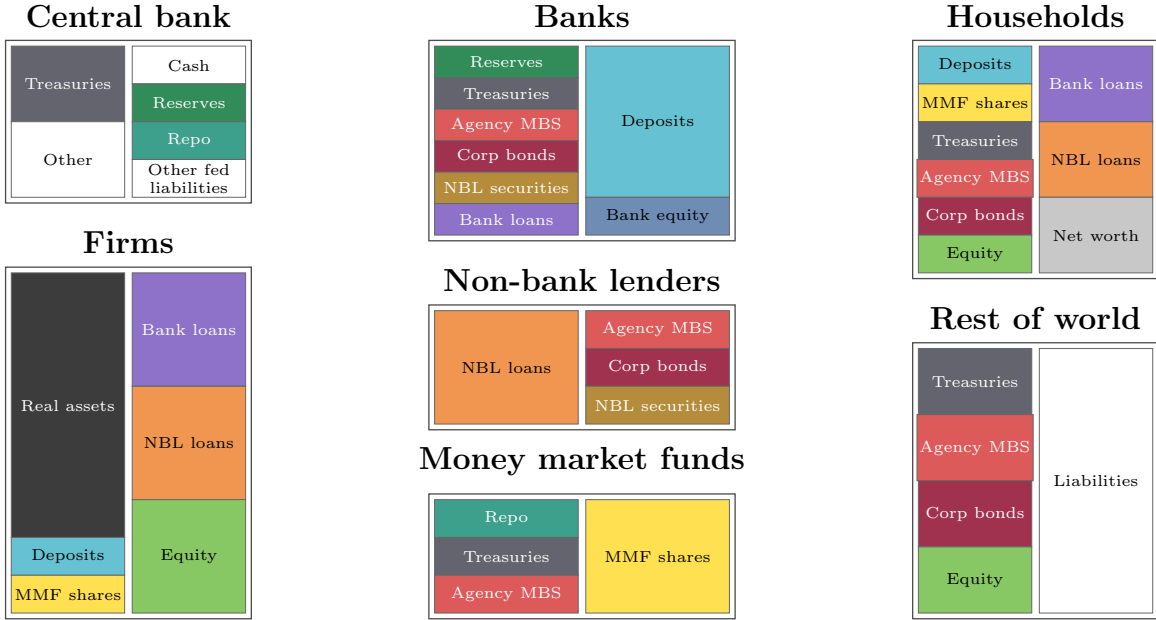
**Intermediaries.** The intermediary sector issues financial instruments to other agents in the economy. It consists of *banks* (**B**), *non-bank liquidity providers* (money market funds, **M**), and *market-based non-bank lenders* (**L**). Banks issue bank loans and invest in securities, while financing their assets with deposits and inside equity, subject to shadow costs of deviating

from target liquidity and capital ratios. Money market funds issue non-deposit liquid savings products and invest in securities and repurchase agreements (repos), earning zero profits in equilibrium. Market-based non-bank lenders make marketable loans to the non-financial sectors which they finance with a composite of bank loans and securities issuance, also under zero-profit. Banks have pricing power in their primary deposit and loan products; MMFs and NBLs are price takers in their funding sources and compete for borrower and saver demand.

**Other sectors.** The rest-of-the-world (RoW) sector consists of all non-domestic savers, who purchase Treasuries, agencies, corporate bonds, and equities according to a CES demand system. The government, representing the consolidated balance sheet of the fiscal and monetary authority, holds an exogenous net position in Treasuries and agencies, and supplies reserves and repo perfectly elastically at policy rates.

**Flow of funds.** Figure 3 illustrates the main flows of funds in the model. Real-sector demand for differentiated savings and borrowing products on the left and right of the figure is satisfied by the intermediary sector in the middle, which itself allocates assets and raises funding through a network of cross-claims.

**Sequencing.** We develop the household and firm sectors first, in Sections 3 and 4, because they constitute the principal empirical contribution of the paper: a structural characterization of how saving and borrowing preferences vary across observable real-sector characteristics. Section 5 then develops the intermediary sector — banks, money market funds, and non-bank lenders — including the bank pricing model that, together with cross-class household and firm demand, governs equilibrium credit conditions. Section 6 closes the model with the government, the rest of the world, and the equilibrium definition.



**Figure 3: Sector balance sheets in the model.** Each sector is shown as a stylized balance sheet with assets on the left and liabilities on the right; within each side, blocks have equal sizes so the figure conveys composition rather than magnitudes. Color encodes the type of financial claim: bank deposits (teal), bank loans to households and firms (purple), MMF shares (yellow), non-bank loans (orange), Treasuries (gray), agency MBS (red), corporate bonds (burgundy), firm equity (green), real assets (dark gray), reserves (dark green), repo (teal-green), bank inside equity (blue-gray), and household net worth or rest-of-world residual liabilities (light gray). The olive color marks bank loans to non-bank lenders, which non-bank lenders issue as a distinct claim and banks hold as part of their liquidity composite. White-filled blocks in the central bank’s balance sheet (*Other*, *Cash*, *Other fed liabilities*) represent items outside the scope of the model.

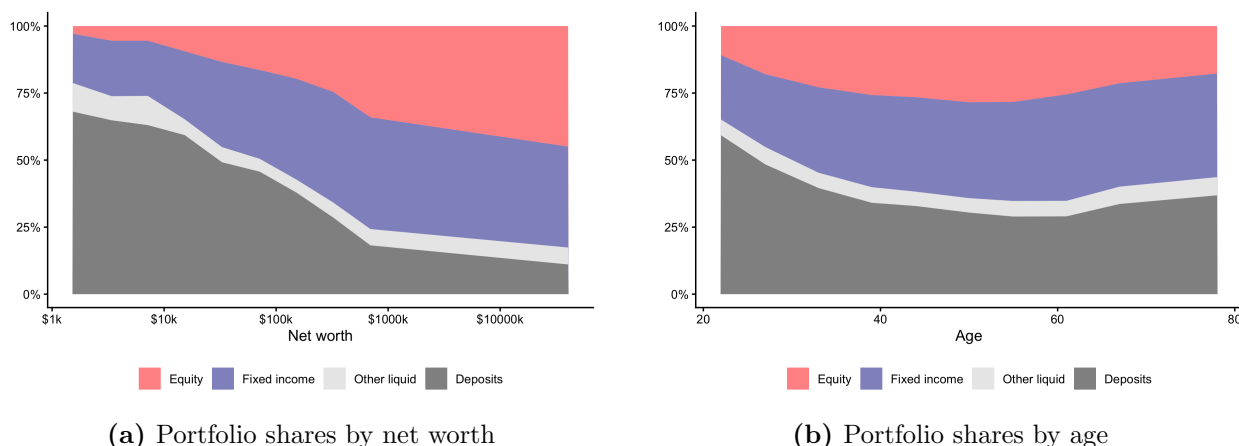
### 3 Household Sector

We start with the household sector. We first document how household saving and borrowing decisions vary with observable characteristics in U.S. microdata. These facts discipline the household side of the model: they show which characteristics matter for the composition of household portfolios, the scale of household borrowing, and the allocation of borrowing across bank and nonbank lenders. Motivated by these patterns, we then specify and estimate a household demand system in which observable household characteristics affect savings and borrowing demand across financial instruments.

#### 3.1 US Micro-Evidence

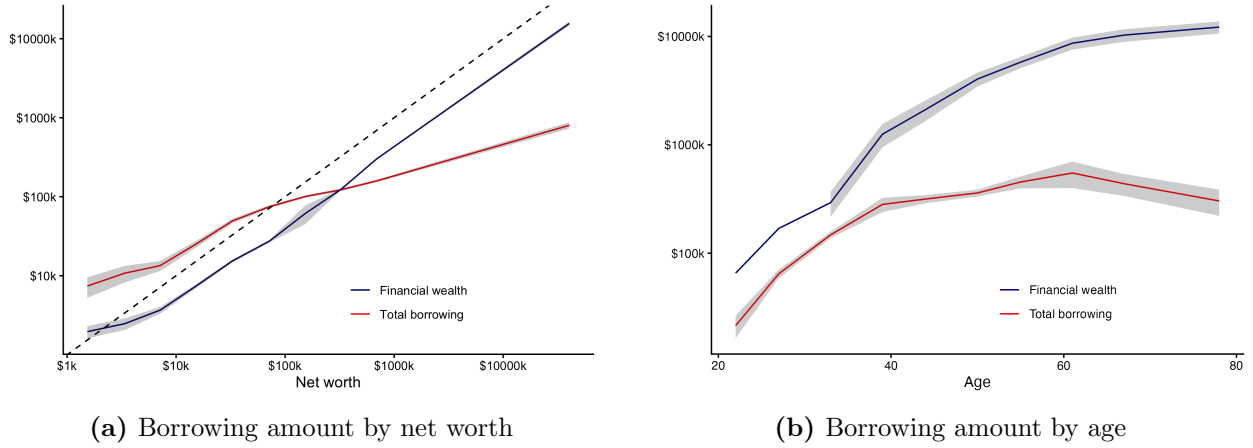
There are two key features of the data that motivate the modeling choices that follow. First, on the saving side, household portfolios vary systematically with both net worth

and age. Panel A of Figure 4 shows that deposits account for a large share of financial wealth among low-net-worth households but decline sharply with wealth, as richer households substitute toward fixed-income and equity claims. Panel B shows that portfolio composition also varies over the life cycle: deposit shares are highest among young households, fall through middle age as fixed-income and equity shares rise, and increase modestly again at older ages. These patterns motivate a household saving problem in which observable household characteristics shift demand across deposits, fixed-income claims, and equity.



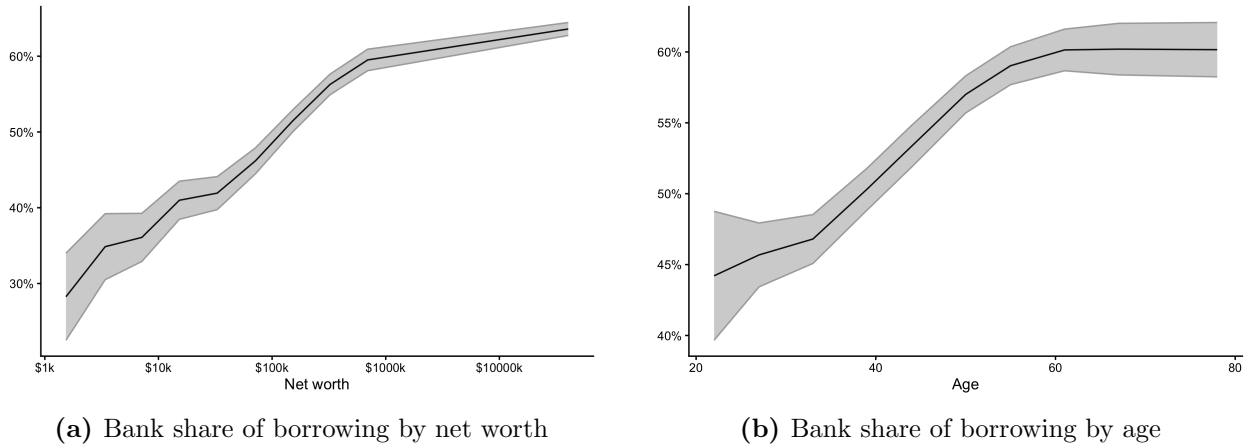
**Figure 4: Household Portfolio Composition by Net Worth and Age.** Panel (a) shows stacked portfolio shares by net worth on a log scale. Panel (b) shows portfolio shares by age. Source: Survey of Consumer Finances.

Second, on the borrowing side, household characteristics first shape *who borrows*. Figure 5 shows that total household borrowing rises with net worth, even though financial leverage falls. Wealthier households therefore borrow more in levels, but they do so from larger balance sheets and with lower leverage. Borrowing also varies over the life cycle: it rises through early and middle age and then declines at older ages, while financial assets continue to rise. Net worth and age therefore jointly shape the scale of household credit demand.



**Figure 5: Household Borrowing: Quantity Margin.** Panel (a) plots average total borrowing (red) and financial assets (blue) against net worth, with the 45-degree line (dashed). Panel (b) shows borrowing and financial assets by age. Source: Survey of Consumer Finances.

Household characteristics also shape *where households borrow*. Conditional on borrowing, Figure 6 shows that the bank share of household borrowing rises sharply with net worth, from roughly 30% at the bottom of the wealth distribution to more than 60% at the top. The bank share also rises over the life cycle, increasing from about 45% among young households to around 60% by late middle age, where it remains roughly flat. Thus, wealth and age affect not only the amount of household borrowing but also its allocation across lender types. Net worth also predicts opposite patterns of bank reliance on the two sides of the household balance sheet: wealthier households rely less on banks as savers, but more on banks as borrowers.



**Figure 6: Bank Share of Household Borrowing.** This figure shows the bank share of total household borrowing by net worth (Panel A) and by age (Panel B). Source: Survey of Consumer Finances.

Table 2 shows that these patterns are robust in a regression setting. On the saving side, deposit shares decline strongly with net worth: after adding financial risk-taking, financial sophistication, saving motives, and year fixed effects, a one-log-point increase in net worth is associated with a 6.7 percentage point lower deposit share. On the borrowing side, conditional on borrower characteristics and year fixed effects, a one-log-point increase in net worth is associated with a 4.5 percentage point higher bank share of borrowing. Age also matters, especially for borrowing, where elderly households are substantially more likely to borrow from banks. These estimates motivate a household problem in which savings and borrowing choices are modeled separately and allowed to vary with household characteristics.

**Table 2: Household Deposit Shares and Borrowing Regressions.** Columns (1)–(2) report regressions of the deposit share of household financial assets; columns (3)–(4) report regressions of the bank share of household borrowing. \* $p < 0.1$ , \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ . Source: Survey of Consumer Finances.

|                           | Deposit share        |                      | Bank loan share     |                     |
|---------------------------|----------------------|----------------------|---------------------|---------------------|
|                           | (1)                  | (2)                  | (3)                 | (4)                 |
| log(Net worth)            | −0.091***<br>(0.001) | −0.067***<br>(0.002) | 0.044***<br>(0.002) | 0.045***<br>(0.003) |
| Middle-aged               | −0.014***<br>(0.004) | −0.041***<br>(0.009) | 0.014**<br>(0.007)  | 0.004<br>(0.013)    |
| Elderly                   | 0.059***<br>(0.005)  | −0.004<br>(0.010)    | 0.076***<br>(0.009) | 0.090***<br>(0.017) |
| Takes financial risk      |                      | −0.086***<br>(0.006) |                     | −0.025**<br>(0.010) |
| Financially sophisticated |                      | −0.060***<br>(0.007) |                     | −0.005<br>(0.010)   |
| Saving for consumption    |                      | 0.019**<br>(0.009)   |                     |                     |
| Saving for other          |                      | 0.042***<br>(0.009)  |                     |                     |
| Borrowing for housing     |                      |                      |                     | −0.012<br>(0.011)   |
| Year FE                   | Yes                  | Yes                  | Yes                 | Yes                 |
| Observations              | 46,024               | 13,285               | 30,438              | 8,565               |
| R <sup>2</sup>            | 0.190                | 0.186                | 0.042               | 0.040               |

### 3.2 Model Setup: Household Problem

The facts above imply three requirements for the household block. First, the saving side must allow households with different net worth and age to demand different portfolios of deposits, fixed-income claims, and equity. Second, the borrowing side must be modeled separately from the saving side, because the same household characteristic—net worth—has opposite implications for bank reliance as a saver and as a borrower. Third, the model must distinguish the quantity of borrowing from the allocation of borrowing across bank and nonbank lenders.

We capture these margins with two connected demand blocks. On the saving side, households allocate financial wealth across deposits, fixed-income claims, and equity. On the borrowing side, a cash-flow constraint pins down total borrowing by limiting interest payments relative to the household’s ability to pay; conditional on this total borrowing, households allocate debt across bank and nonbank credit. Observable household characteristics shift the across-class preference weights in both the saving and borrowing blocks, while returns determine substitution across instruments. Inner nests allocate deposits and bank loans across individual banks, which govern bank pricing and equilibrium computation, but are not identified from the household-level micro moments used in this section.

**Individual household’s problem.** Households are heterogeneous in their preferences and net worth. Household  $i$  derives utility over consumption and differentiated returns from financial classes indexed by  $j \in J_S^h$ :

$$u^i = u^i(c^i, \mathcal{S}^i) \quad \mathcal{S}^i \equiv \left( \sum_{j \in J_S^h} \alpha_j^i (r_j q_{s,j}^i)^{\frac{\sigma_S^h - 1}{\sigma_S^h}} \right)^{\frac{\sigma_S^h}{\sigma_S^h - 1}}. \quad (1)$$

Here,  $c^i$  is household  $i$ ’s current consumption, and  $\mathcal{S}^i$  is an aggregator over differentiated class returns that depends on  $i$ -specific utility weights  $\alpha_j^i$ , class holdings  $q_{s,j}^i$ , class returns  $r_j$ , and elasticity of substitution across savings classes  $\sigma_S^h$ . For non-bank classes (Treasuries, agency MBS, corporate bonds, equity, MMF shares), each class is a homogeneous market claim with a single return  $r_j$ . For the deposit class,  $q_{s,\text{dep}}^i$  is itself a CES aggregator over

individual banks  $k \in K_h$ , where banks are differentiated but symmetric:

$$q_{s,\text{dep}}^i \equiv \left( \sum_{k \in K_h} (q_{s,\text{dep},k}^i)^{\frac{\sigma_{D,h}^h - 1}{\sigma_{D,h}^h}} \right)^{\frac{\sigma_{D,h}^h}{\sigma_{D,h}^h - 1}}, \quad (2)$$

where  $q_{s,\text{dep},k}^i$  is household  $i$ 's deposit holding at bank  $k$  and  $\sigma_{D,h}^h$  is the within-class elasticity of substitution across banks for deposits.

The household finances consumption and savings out of net worth  $w^i$  and wealth-equivalent borrowing  $\mathcal{B}^i$ .  $\mathcal{B}^i$  is a CES aggregator over bank loans and non-bank loans, indexed by  $j \in J_B^h \equiv \{b, n\}$ . The budget constraint and wealth-equivalent borrowing are as follows:

$$w^i + \mathcal{B}^i \geq \sum_{j \in J_S^h} q_{s,j}^i + c^i \quad \mathcal{B}^i \equiv \left( \sum_{j \in J_B^h} \beta_j^i (q_{b,j}^i)^{\frac{\sigma_B^h - 1}{\sigma_B^h}} \right)^{\frac{\sigma_B^h}{\sigma_B^h - 1}}. \quad (3)$$

$w^i$  is net worth,  $\beta_j^i$  is  $i$ -specific utility weights on borrowing class  $j$ ,  $q_{b,j}^i$  is the dollar borrowing position in borrowing class  $j$ , and  $\sigma_B^h$  is the elasticity of substitution across borrowing classes. Analogous to the deposit class, the bank-loan class is itself a CES aggregator over individual banks, with banks differentiated but symmetric:

$$q_{b,b}^i \equiv \left( \sum_{k \in K_h} (q_{b,b,k}^i)^{\frac{\sigma_{L,h}^h - 1}{\sigma_{L,h}^h}} \right)^{\frac{\sigma_{L,h}^h}{\sigma_{L,h}^h - 1}}, \quad (4)$$

where  $\sigma_{L,h}^h$  is the within-class elasticity of substitution across banks for loans. Total borrowing is limited by a cashflow constraint, where total interest payments on borrowing must not exceed the household's ability to pay,  $y^i$ :

$$y^i \geq \sum_{j \in J_B^h} b_j^i r_j. \quad (5)$$

For notational convenience, we denote the total interest payment as  $D^i \equiv \sum_{j \in J_B^h} b_j^i r_j$ , and the total dollars spent on savings classes  $A^i \equiv \sum_{j \in J_S^h} a_j^i$ .

The household's problem is to maximize utility by choosing consumption, savings classes, and borrowing classes, subject to the budget constraint (3) and cashflow constraint (5). The

problem decomposes into a two-stage allocation: across classes (the outer CES nest) and across banks within the deposit and bank-loan classes (the inner CES nests). Taking effective savings quantity  $\mathcal{S}^i$  and borrowing quantity  $\mathcal{B}^i$  as given, individual class quantities and class demands are as follows:

$$q_{s,j}^i r_j = \mathcal{S}^i (\alpha_j^i)^{\sigma_S^h} \left( \frac{r_j}{R_s^i} \right)^{\sigma_S^h}, \quad R_s^i = \left( \sum_{j \in J_S^h} (\alpha_j^i)^{\sigma_S^h} r_j^{\sigma_S^h - 1} \right)^{\frac{1}{\sigma_S^h - 1}}, \quad \mathcal{S}^i = A^i R_s^i \quad (6)$$

$$q_{b,j}^i = \mathcal{B}^i (\beta_j^i)^{\sigma_B^h} \left( \frac{r_j}{R_b^i} \right)^{-\sigma_B^h}, \quad R_b^i = \left( \sum_{j \in J_B^h} (\beta_j^i)^{\sigma_B^h} r_j^{1 - \sigma_B^h} \right)^{\frac{1}{1 - \sigma_B^h}}, \quad \mathcal{B}^i = D^i / R_b^i. \quad (7)$$

For the deposit and bank-loan classes, the class-level returns  $r_{\text{dep}}$  and  $r_b$  entering (6)–(7) are themselves CES aggregates of individual bank rates, and class-level quantities allocate across banks according to within-class CES demand:

$$r_{\text{dep}} = \left( \sum_{k \in K_h} r_{\text{dep},k}^{\sigma_{D,h}^h - 1} \right)^{\frac{1}{\sigma_{D,h}^h - 1}}, \quad q_{s,\text{dep},k}^i r_{\text{dep},k} = q_{s,\text{dep}}^i r_{\text{dep}} \left( \frac{r_{\text{dep},k}}{r_{\text{dep}}} \right)^{\sigma_{D,h}^h}, \quad (8)$$

$$r_b = \left( \sum_{k \in K_h} r_{b,k}^{1 - \sigma_{L,h}^h} \right)^{\frac{1}{1 - \sigma_{L,h}^h}}, \quad q_{b,b,k}^i = q_{b,b}^i \left( \frac{r_{b,k}}{r_b} \right)^{-\sigma_{L,h}^h}. \quad (9)$$

Finally, specifying that,

$$u^i(c^i, \mathcal{S}^i) = \left( \gamma_c^i (c^i)^{\frac{\sigma^h - 1}{\sigma^h}} + \gamma_s^i (\mathcal{S}^i)^{\frac{\sigma^h - 1}{\sigma^h}} \right)^{\frac{\sigma^h}{\sigma^h - 1}}, \quad (10)$$

and using the budget constraint  $c^i + \mathcal{S} / R_s^i = w^i + \mathcal{B}^i$ , one obtains

$$\mathcal{S}^i = \gamma_s^i (w^i + \mathcal{B}^i) R_s^i \left( \frac{R_s^i}{R^i} \right)^{\sigma^h}, \quad R^i = \left( (\gamma_c^i)^{\sigma^h} + (\gamma_s^i)^{\sigma^h} (R^i)^{\sigma^h - 1} \right)^{\frac{1}{\sigma^h - 1}}. \quad (11)$$

Finally,  $\mathcal{B}^i$  is pinned down by the cashflow constraint,  $\mathcal{B}^i R_b^i = y^i$ .

**Aggregation.** Total household demand at the class level for savings class  $j \in J_S^h$  and for borrowing class  $j \in J_B^h$  are obtained by aggregating over all households:

$$Q_j^s = \sum_i q_{s,j}^i, \quad Q_j^b = \sum_i q_{b,j}^i. \quad (12)$$

For the deposit and bank-loan classes, bank-level aggregates are obtained similarly:

$$Q_{\text{dep},k}^s = \sum_i q_{s,\text{dep},k}^i \quad Q_{b,k}^b = \sum_i q_{b,b,k}^i. \quad (13)$$

Under the assumption that households face national-average bank rates within each class — that is,  $r_{\text{dep},k} = r_{\text{dep}}$  and  $r_{b,k} = r_b$  for all  $k$  in the household-level data — the within-class elasticities  $\sigma_{D,h}^h$  and  $\sigma_{L,h}^h$  do not enter the across-class household demand system (6)–(7). They are model primitives identified separately from bank rate passthrough rather than from household-level data. The remaining household-side parameters are the across-class elasticities and the characteristic-dependent preference weights. We now estimate these objects using the same division suggested by the facts: aggregate time-series variation identifies substitution elasticities, while SCF cross-sectional variation identifies how net worth and age shift demand across instruments.

### 3.3 Estimation

The estimation follows the nested structure of the household problem and the empirical patterns documented in Section 3.1. The facts show that net worth and age predict portfolio composition, borrowing amounts, and lender choice. The model interprets these gradients as variation in household demand for different financial instruments. To recover this demand, we separate elasticities from heterogeneity. Aggregate time-series variation in relative returns identifies the across-class elasticities, while SCF microdata identify how the implied preference weights vary with household characteristics.

Estimation proceeds in three stages.

**Stage 1: across-class elasticities from Financial Accounts time series.** The saving-side demand equation (6) implies that, for any two savings classes  $k$  and  $l$ ,

$$\log\left(\frac{A_{k,t}}{A_{l,t}}\right) = \sigma_S^h \log\left(\frac{\bar{\alpha}_k}{\bar{\alpha}_l}\right) + (\sigma_S^h - 1) \log\left(\frac{r_{k,t}}{r_{l,t}}\right) + \tilde{\varepsilon}_{k,l,t}, \quad (14)$$

where  $A_{k,t}$  is aggregate household holdings of asset class  $k$  at quarter  $t$ ,  $r_{k,t}$  is the population-average return on class  $k$ , and  $\bar{\alpha}_k$  is the population-average preference weight for class  $k$ .<sup>1</sup>

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<sup>1</sup>At the aggregate level, the within-class price index for deposits collapses to the average deposit return when within-class preference weights are not parameterized as functions of household characteristics. The within-class elasticities therefore do not enter equation (14).

We estimate this equation using quarterly Financial Accounts data on household holdings of deposits, MMF shares, Treasuries, agencies, corporate bonds, and equities. Asset-class fixed effects absorb the average preference-weight ratios. Identification of  $\sigma_S^h$  comes from time-series variation in relative returns across savings classes. We instrument the relative return with the federal funds rate interacted with asset-class indicators and with the Federal Reserve’s Treasury holdings, which generate variation in cross-asset spreads.

The borrowing-side analog follows from equation (7). For bank and nonbank credit classes  $k$  and  $l$ ,

$$\log\left(\frac{\mathcal{B}_{k,t}}{\mathcal{B}_{l,t}}\right) = \sigma_B^h \log\left(\frac{\bar{\beta}_k}{\bar{\beta}_l}\right) - \sigma_B^h \log\left(\frac{r_{k,t}^B}{r_{l,t}^B}\right) + \tilde{\varepsilon}_{k,l,t}, \quad (15)$$

where  $\mathcal{B}_{k,t}$  is aggregate household borrowing in class  $k$  and  $r_{k,t}^B$  is the corresponding borrowing rate. We estimate this equation using quarterly Financial Accounts data on the bank-versus-nonbank composition of household credit, instrumenting the relative borrowing rate with the federal funds rate and Federal Reserve Treasury holdings. The results are summarized in Table 3.

**Table 3: Household Cross-Asset Substitution Estimation.** This table shows the estimates for the household cross-asset substitution parameter,  $\sigma_B^h$  and  $\sigma_S^h$ , estimated using time-series variation in aggregate savings and borrowing quantities and interest rates. Columns (1) and (2) show the results for savers. The estimated coefficient maps to the structural parameter as  $\sigma_S^h - 1$ . Columns (3) and (4) show the results for borrowers. The estimated coefficient maps to the structural parameter as  $-\sigma_B^h$ . Columns (1) and (3) are OLS estimates. Columns (2) and (4) instrument for rates using the fed funds rate, interacted with the asset type for the saver estimate. Standard errors are shown in parentheses. \*p<0.1; \*\*p<0.05; \*\*\*p<0.01.

|                    | <i>Dependent variable: log S<sub>j</sub>/S<sub>0</sub></i> |                     |                    |                      |
|--------------------|------------------------------------------------------------|---------------------|--------------------|----------------------|
|                    | Saving                                                     |                     | Borrowing          |                      |
|                    | $\sigma_S^h - 1$                                           |                     | $-\sigma_B^h$      |                      |
|                    | (1)                                                        | (2)                 | (3)                | (4)                  |
| log $P_j/P_0$      | 0.159***<br>(0.036)                                        |                     | -0.384*<br>(0.210) |                      |
| log $P_j/P_0$ (IV) |                                                            | 0.291***<br>(0.049) |                    | -1.547***<br>(0.462) |
| Observations       | 200                                                        | 200                 | 40                 | 40                   |
| R <sup>2</sup>     | 0.802                                                      | 0.788               | 0.623              | 0.309                |

**Stage 2: preference heterogeneity from SCF microdata.** Given  $\sigma_S^h$  from Stage 1, the household saving demand equation (6) can be rearranged as

$$\log\left(\frac{A_{k,t}^i}{A_{l,t}^i}\right) - (\sigma_S^h - 1) \log\left(\frac{r_{k,t}}{r_{l,t}}\right) = \sigma_S^h \log\left(\frac{\alpha_k^i}{\alpha_l^i}\right) + \tilde{\varepsilon}_{k,l,t}^i. \quad (16)$$

We parameterize the relative preference weights as functions of observable household characteristics,

$$\log\left(\frac{\alpha_k^i}{\alpha_l^i}\right) = \alpha_{k,l}^0 + \alpha_{k,l}^w \log w^i + \alpha_{k,l}^a \text{Age}^i + X_i' \alpha_{k,l}^X, \quad (17)$$

where  $X_i$  includes the additional household controls used in the reduced-form evidence. With  $\sigma_S^h$  taken as given and returns measured at the aggregate class level, the left-hand side of equation (16) is observed in the SCF. OLS estimates equation (17), recovering how net worth, age, and other household characteristics shift demand across savings instruments.

The borrowing side proceeds analogously. Given  $\sigma_B^h$  from Stage 1, equation (7) implies

$$\log\left(\frac{\mathcal{B}_{k,t}^i}{\mathcal{B}_{l,t}^i}\right) + \sigma_B^h \log\left(\frac{r_{k,t}^B}{r_{l,t}^B}\right) = \sigma_B^h \log\left(\frac{\beta_k^i}{\beta_l^i}\right) + \tilde{\varepsilon}_{k,l,t}^i. \quad (18)$$

We parameterize relative borrowing weights as

$$\log\left(\frac{\beta_k^i}{\beta_l^i}\right) = \beta_{k,l}^0 + \beta_{k,l}^w \log w^i + \beta_{k,l}^a \text{Age}^i + X_i' \beta_{k,l}^X. \quad (19)$$

OLS estimates equation (19), identifying how the preference for bank versus nonbank borrowing varies with household characteristics. This step uses the lender type-choice patterns documented in Section 3.1 to estimate how household characteristics shift the relative demand for bank versus nonbank borrowing.

**Stage 3: consumption versus savings and borrowing capacity.** After estimating the across-class saving weights, we construct the household-specific savings return index  $R_s^i$  from equation (6). The outer saving nest in equation (11) implies the estimating equation

$$\log\left(\frac{\mathcal{S}^i}{c^i}\right) = \log\left(\frac{\gamma_s^i}{\gamma_c^i}\right) + \sigma^h \log R_s^i + \tilde{\varepsilon}^i, \quad (20)$$

where  $\mathcal{S}^i$  is the savings aggregate and  $c^i$  is consumption. The coefficient on  $\log R_s^i$  identifies the elasticity between consumption and the savings aggregate,  $\sigma^h$ , while household characteristics identify heterogeneity in  $\gamma_s^i/\gamma_c^i$ . We instrument  $\log R_s^i$  with the federal funds rate interacted with household net worth.

Finally, the borrowing quantity is disciplined by the cash-flow constraint in equation (5). In the model, this constraint pins down total borrowing through  $\mathcal{B}^i R_b^i = y^i$ , where  $R_b^i$  is the borrowing price index from equation (7). We estimate the empirical counterpart by relating observed debt-service capacity to household characteristics in the SCF. This step disciplines the scale of household borrowing, while Stage 2 disciplines the allocation of that borrowing across bank and nonbank lenders.

**Identification, summarized.** The nested structure produces a clean division of identification across data sources. Across-class elasticities  $\sigma_S^h$  and  $\sigma_B^h$  are identified from time-series variation in relative returns in quarterly Financial Accounts data. Preference heterogeneity  $(\alpha^w, \alpha^a, \beta^w, \beta^a)$  is identified from cross-sectional variation in SCF microdata, given the across-class elasticities. The outer-nest elasticity  $\sigma^h$  is identified from variation in the savings return index, instrumented by monetary-policy-induced shifts in returns. The within-class elasticities  $\sigma_{D,h}^h$  and  $\sigma_{L,h}^h$  do not enter this section’s estimation because SCF moments use aggregate class returns rather than household-specific within-class CES price indexes. These elasticities instead govern how individual banks compete for household deposits and loans and are estimated with the intermediary-side parameters in Section 5.

The results are reported in Table 4.

**Table 4: Household CES Demand Estimation.** Column (1) estimates the savings-nest CES, where the dependent variable is the log portfolio share of asset  $j$  relative to deposits ( $\log \alpha_j / \alpha_{\text{dep}}$ ). Regressors include log net worth, age dummies, asset-type fixed effects, and interactions. Column (2) estimates the borrowing-nest CES, where the dependent variable is the log ratio of bank to nonbank borrowing preference ( $\log \beta_b / \beta_{nb}$ ). Columns (3)–(4) estimate the outer-nest elasticity between consumption and savings, where the dependent variable is  $\log(c/\mathcal{S})$ ; column (3) is OLS and column (4) instruments for the return index  $R_s$  with asset returns.

|                    | <i>Dependent variable:</i>                             |                                           |                                      |                                     |
|--------------------|--------------------------------------------------------|-------------------------------------------|--------------------------------------|-------------------------------------|
|                    | $\log(\alpha_j / \alpha_{\text{dep}})$<br>Savings Nest | $\log(\beta_b / \beta_{nb})$<br>Debt Nest | $\log(c/\mathcal{S})$<br>Outer (OLS) | $\log(c/\mathcal{S})$<br>Outer (IV) |
|                    | (1)                                                    | (2)                                       | (3)                                  | (4)                                 |
| log(net worth)     | 0.424***<br>(0.015)                                    | 0.061***<br>(0.007)                       | 0.332***<br>(0.053)                  | 0.328***<br>(0.053)                 |
| Middle-aged        | 0.003<br>(0.074)                                       | −0.001<br>(0.034)                         | −0.647***<br>(0.139)                 | −0.632***<br>(0.140)                |
| Elderly            | 0.019<br>(0.090)                                       | 0.071*<br>(0.040)                         | 0.203<br>(0.304)                     | 0.251<br>(0.306)                    |
| MMF                | −2.375***<br>(0.064)                                   |                                           |                                      |                                     |
| Treasury           | −1.981***<br>(0.043)                                   |                                           |                                      |                                     |
| Agency             | −2.342***<br>(0.043)                                   |                                           |                                      |                                     |
| Corporate          | −1.475***<br>(0.043)                                   |                                           |                                      |                                     |
| Equity             | −0.896***<br>(0.064)                                   |                                           |                                      |                                     |
| Middle × Bond      | 0.327***<br>(0.086)                                    |                                           |                                      |                                     |
| Elderly × Bond     | 0.152<br>(0.104)                                       |                                           |                                      |                                     |
| Middle × Equity    | 0.129<br>(0.105)                                       |                                           |                                      |                                     |
| Elderly × Equity   | −0.597***<br>(0.127)                                   |                                           |                                      |                                     |
| log $W$ × Bond     | −0.193***<br>(0.018)                                   |                                           |                                      |                                     |
| log $W$ × Equity   | −0.027<br>(0.022)                                      |                                           |                                      |                                     |
| log $R_s$          |                                                        |                                           | 0.980***<br>(0.086)                  |                                     |
| log $R_s$ (fitted) |                                                        |                                           |                                      | 1.046***<br>(0.099)                 |
| Constant           |                                                        | −0.308***<br>(0.029)                      | 1.621***<br>(0.432)                  | 1.936***<br>(0.493)                 |
| Observations       | 3,560                                                  | 712                                       | 297                                  | 297                                 |
| R <sup>2</sup>     | 0.556                                                  | 0.128                                     | 0.420                                | 0.419                               |

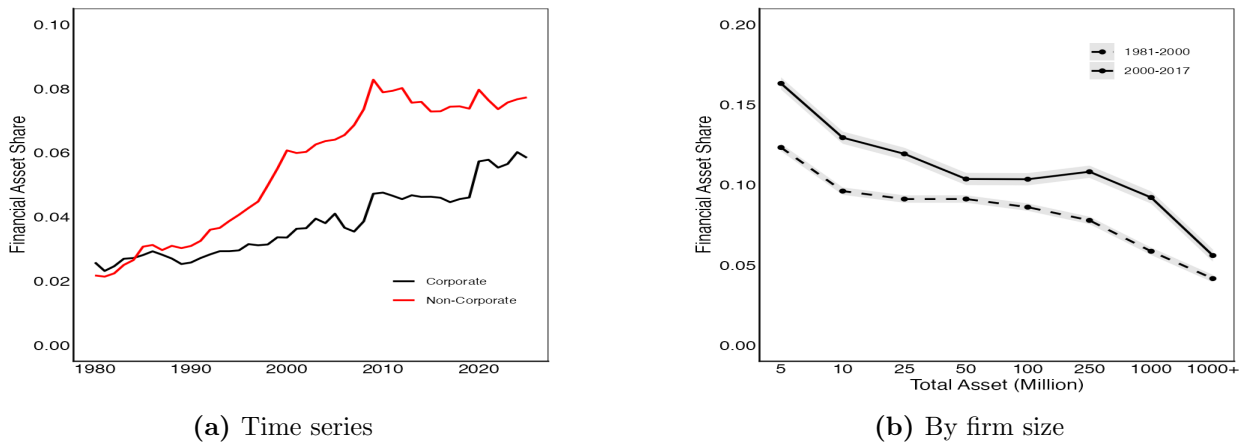
*Note:* \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

## 4 Firms

We turn next to the firm sector. We first document how firm balance-sheet choices vary with observable characteristics in U.S. microdata. These facts discipline the firm side of the model: they show which firm characteristics matter for the allocation of assets between real and financial claims, the composition of financial assets between deposits and market-based instruments, the scale of borrowing, and the allocation of debt across bank and nonbank credit. Motivated by these patterns, we then specify and estimate a firm demand system in which firm size and industry affect asset and liability demand across financial instruments.

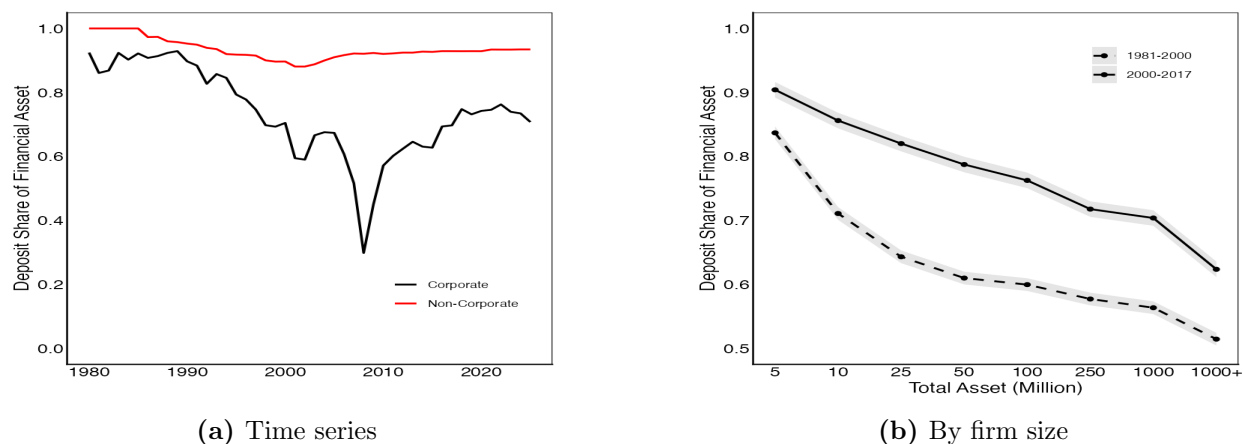
### 4.1 US Micro-Evidence

There are two key features of the data that motivate the modeling choices that follow. First, on the asset side, firms differ in the share of assets held in financial form, and this margin varies both over time and across the firm-size distribution. Panel A of Figure 7 shows that financial assets have become a larger share of nonfinancial corporate assets over time, rising from roughly 4% of total assets in 1980 to about 8% by 2020. One interpretation of this aggregate increase is a changing composition of the firm sector, including the rise of more intangible-intensive and service-oriented industries. Panel B shows that financial-asset shares are also systematically related to size: smaller firms hold larger financial buffers than larger firms. These patterns motivate a firm asset problem in which both industry and size shift the allocation of total assets between real assets and a financial-asset composite.



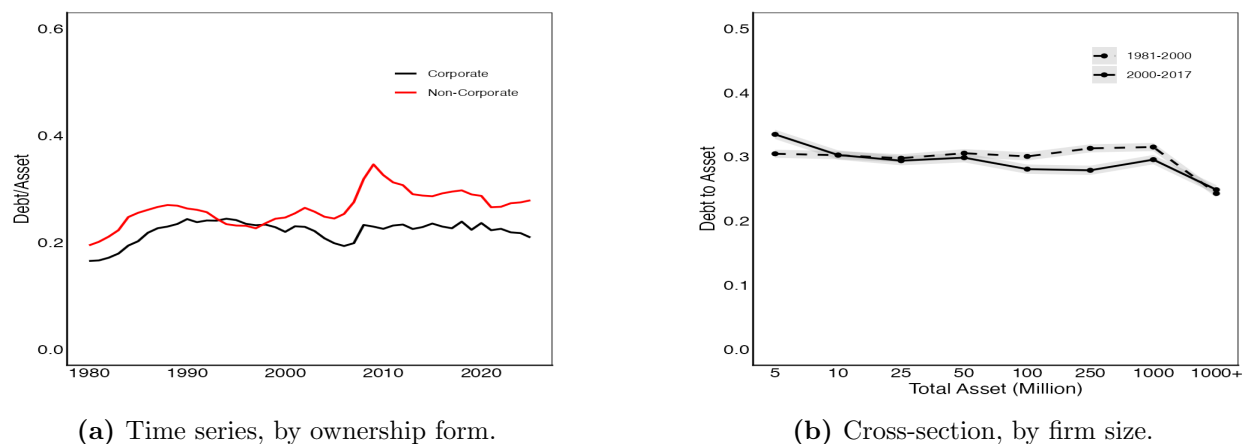
**Figure 7: Firm Financial Asset Share.** Panel (a) shows financial assets as a share of total assets for corporate (black) and noncorporate (red) nonfinancial businesses, 1980–2020. Panel (b) shows the financial-asset share by firm size class, comparing 2000–2017 (solid) with 1981–2000 (dashed), with 95% confidence bands. Sources: Financial Accounts of the United States and Census Quarterly Financial Report.

The composition of financial assets also varies systematically. Panel A of Figure 8 shows that the deposit share of corporate financial assets has declined from above 90% in 1980 to roughly 70% by 2020, while noncorporate firms remain much more deposit-intensive. Panel B shows that smaller firms hold a larger share of financial assets as deposits, whereas larger firms substitute toward MMFs and other market-based instruments. These patterns motivate a financial-asset nest in which firms allocate liquidity across deposits and market-based claims.



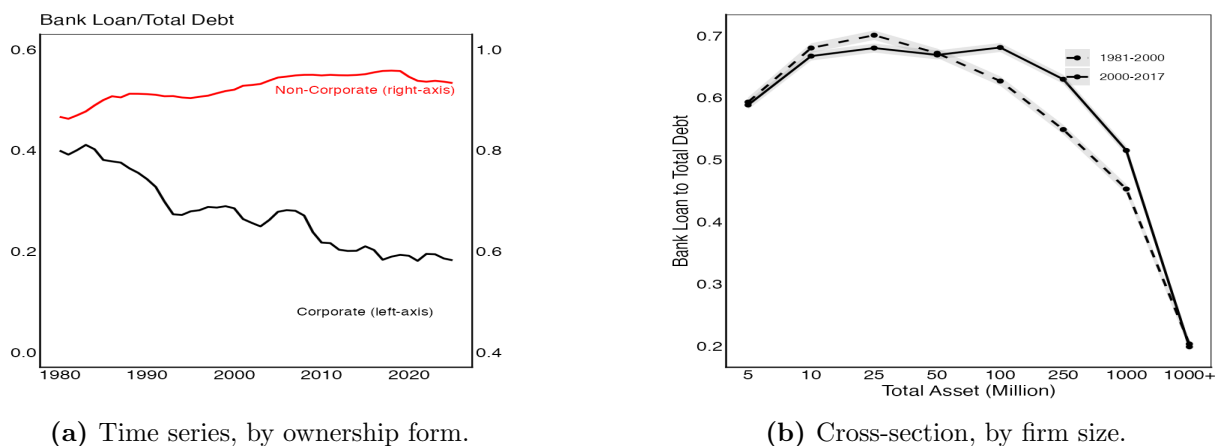
**Figure 8: Deposit Share of Firm Financial Assets.** Panel (a) shows deposits as a share of financial assets for corporate (black) and noncorporate (red) nonfinancial businesses, 1980–2020. Panel (b) shows the deposit share of financial assets by firm size class, comparing 2000–2017 (solid) with 1981–2000 (dashed), with 95% confidence bands. Sources: Financial Accounts of the United States and Census Quarterly Financial Report.

Second, on the liability side, firm characteristics shape both how much firms borrow and where they borrow. Figure 9 shows that aggregate corporate leverage has been relatively stable over time, with debt equal to roughly 20–30% of total assets. At the same time, leverage varies substantially across the firm-size distribution. In the post-2000 sample, smaller firms are more levered than larger firms, reversing the size profile observed earlier in the sample. These patterns motivate a liability side in which firms’ financing scale is linked to balance-sheet conditions and financing frictions that can vary across firm types.



**Figure 9: Firm Leverage (Debt-to-Asset Ratio).** Panel (a) shows total interest-bearing debt as a share of total assets for corporate (black) and noncorporate (red) nonfinancial businesses, 1980–2020. Panel (b) shows the debt-to-asset ratio by firm size class, comparing 2000–2017 (solid) with 1981–2000 (dashed), with 95% confidence bands. Sources: Financial Accounts of the United States and Census Quarterly Financial Report.

Firm characteristics also shape the composition of debt. Panel A of Figure 10 shows that the bank-loan share of corporate debt has declined from roughly 40% in 1980 to about 20% by 2020, while noncorporate firms remain much more bank-dependent. Panel B shows that bank reliance is non-monotone in firm size: the bank-loan share is highest among mid-sized firms and substantially lower among the largest firms, which rely more heavily on market-based debt. These patterns motivate a firm borrowing problem in which firm characteristics shift demand across bank and nonbank credit.



**Figure 10: Bank-Loan Share of Firm Debt.** Panel (a) shows bank loans as a share of total debt for corporate (black) and noncorporate (red) nonfinancial businesses, 1980–2020. Panel (b) shows the bank-loan share of total debt by firm size class, comparing 2000–2017 (solid) with 1981–2000 (dashed), with 95% confidence bands. Sources: Financial Accounts of the United States and Census Quarterly Financial Report.

## 4.2 Model Setup: Firm Problem

The facts above imply three requirements for the firm block. First, the asset side must allow firms to choose the scale of financial assets relative to real assets, with this allocation varying across firm types. Second, the financial-asset block must distinguish deposits from market-based liquidity, since smaller firms are more deposit intensive while larger firms rely more on MMFs and other market instruments. Third, the liability side must distinguish the scale of borrowing from the composition of borrowing, allowing firm size and industry to shift demand between bank and nonbank debt.

We capture these margins with a nested demand system on both sides of the corporate balance sheet. On the asset side, firms allocate total assets between real assets and a financial-asset composite; within the financial composite, they allocate liquidity between deposits and MMFs. On the liability side, firms finance assets with equity and a debt composite; within the debt composite, they allocate borrowing between bank and nonbank credit. Observable firm characteristics shift the across-class weights in these nests, while relative returns determine substitution across instruments. Inner nests allocate deposits and bank loans across individual banks. These within-class nests govern bank pricing and equilibrium computation, but are not identified from the firm-level moments used in this section.

Firm scale is pinned down separately. We assume free mobility of capital across firms and industries, so aggregate firm scale adjusts until the value-weighted average Tobin's  $q$  equals one. Decreasing returns to real projects then determine how changes in required returns translate into real investment. We first describe the individual firm's portfolio and financing choices, and then describe aggregation and scale determination.

**Individual firm's problem.** The firm seeks to maximize cash flows to equity holders, subject to the balance-sheet constraint that assets equal liabilities:

$$\Pi^i = R_A^i A^i - R_D^i \mathcal{D}^i - \Phi^i(E^i; A^i), \quad A^i = \mathcal{D}^i + E^i. \quad (21)$$

Here,  $R_A^i$  is the return on assets,  $A^i$  is total firm assets,  $R_D^i$  is the cost of debt,  $\mathcal{D}^i$  is the effective funding capacity of the firm's debt,  $E^i$  is firm equity, and  $\Phi^i(E^i; A^i)$  is the cost of deviating from the target equity-to-assets ratio. This cost captures, in reduced form,

financial frictions that affect equity financing. We impose

$$\Phi^i(E^i; A^i) = \frac{\eta_E^i}{2} \left( \frac{E^i}{A^i} - \bar{s}^i \right)^2 A^i. \quad (22)$$

Taking  $R_A^i$ ,  $R_D^i$ , and the composition of  $\mathcal{D}^i$  as given for now, imposing the balance-sheet equation into the profit equation yields the optimal equity-to-assets ratio:

$$\left( \frac{E^i}{A^i} \right)^* = \bar{s}^i + \frac{R_D^i}{\eta_E^i}. \quad (23)$$

We pin down firm scale after discussing aggregation.

**Asset side composition.** On the asset side, firm  $i$  allocates total assets  $A^i$  across real capital and a financial composite of deposits and money market funds. The financial portfolio returns are themselves a composite of individual asset holdings: bank deposits and MMF shares. Let  $j \in J_a^F = \{\text{dep}, \text{mmf}\}$  index the types of financial assets. Let  $q_j^i$  denote the dollar holdings of asset  $j$ , and let  $F^i = \sum_{j \in J_a^F} q_j^i$  denote total dollars invested by firm  $i$  in financial assets. Composite financial returns are given by the CES aggregator

$$\mathcal{F}^i = \left( \sum_{j \in J_a^F} \gamma_j^i (r_j q_j^i)^{\frac{\sigma_F^f - 1}{\sigma_F^f}} \right)^{\frac{\sigma_F^f}{\sigma_F^f - 1}}. \quad (24)$$

Here,  $\gamma_j^i$  is the firm- $i$  specific production weight on asset  $j$ ,  $r_j$  is the return on asset  $j$ , and  $\sigma_F^f$  is the elasticity of substitution across financial classes. The deposit class is itself a CES aggregator over individual banks  $k \in K_f$ , where banks are differentiated but symmetric:

$$q_{\text{dep}}^i \equiv \left( \sum_{k \in K_f} (q_{\text{dep},k}^i)^{\frac{\sigma_{D,f}^f - 1}{\sigma_{D,f}^f}} \right)^{\frac{\sigma_{D,f}^f}{\sigma_{D,f}^f - 1}}, \quad (25)$$

where  $q_{\text{dep},k}^i$  is firm  $i$ 's deposit holding at bank  $k$  and  $\sigma_{D,f}^f$  is the within-class elasticity of substitution across banks for firm deposits.

Total returns, inclusive of real and financial returns, are themselves a CES composite,

$\mathcal{A}^i$ :

$$\mathcal{A}^i = \left( \alpha_{\text{real}}^i (r_{\text{real}} q_{\text{real}}^i)^{\frac{\sigma_A^f - 1}{\sigma_A^f}} + \alpha_{\text{fin}}^i (R_{\text{fin}}^i F^i)^{\frac{\sigma_A^f - 1}{\sigma_A^f}} \right)^{\frac{\sigma_A^f}{\sigma_A^f - 1}}. \quad (26)$$

$\alpha_{\text{real}}^i$  and  $\alpha_{\text{fin}}^i$  are  $i$ -specific production weights over real and financial assets, capturing how firm  $i$  uses inputs such as machines, working capital, and liquid assets to earn profits.  $r_{\text{real}}$  is the return on real assets,  $q_{\text{real}}^i$  is the dollar value of real assets,  $R_{\text{fin}}^i$  is the return to the firm-specific financial asset portfolio, and  $\sigma_A^f$  is the elasticity of substitution on the asset side of the firm.

Maximizing total returns for a given scale  $A^i$  yields the following firm-level demand functions and price indices. At the level of the overall nest, the firm-specific price index  $R_A^i$  is

$$R_A^i = \left( (\alpha_{\text{real}}^i)^{\sigma_A^f} r_{\text{real}}^{\sigma_A^f - 1} + (\alpha_{\text{fin}}^i)^{\sigma_A^f} (R_{\text{fin}}^i)^{\sigma_A^f - 1} \right)^{\frac{1}{\sigma_A^f - 1}}, \quad (27)$$

and the optimal choice of real and financial investment is

$$q_{\text{real}}^i = A^i (\alpha_{\text{real}}^i)^{\sigma_A^f} \left( \frac{r_{\text{real}}}{R_A^i} \right)^{\sigma_A^f - 1}, \quad F^i = A^i (\alpha_{\text{fin}}^i)^{\sigma_A^f} \left( \frac{R_{\text{fin}}^i}{R_A^i} \right)^{\sigma_A^f - 1}. \quad (28)$$

Within the financial nest,

$$R_{\text{fin}}^i = \left( \sum_{j \in J_a^F} (\gamma_j^i)^{\sigma_F^f} r_j^{\sigma_F^f - 1} \right)^{\frac{1}{\sigma_F^f - 1}}, \quad q_j^i = F^i (\gamma_j^i)^{\sigma_F^f} \left( \frac{r_j}{R_{\text{fin}}^i} \right)^{\sigma_F^f - 1}. \quad (29)$$

For the deposit class ( $j = \text{dep}$ ), the class-level return  $r_{\text{dep}}$  entering the within-financial demand above is itself a CES aggregate of individual bank deposit rates, and class-level quantity allocates across banks according to within-class CES demand:

$$r_{\text{dep}} = \left( \sum_{k \in K_f} r_{\text{dep},k}^{\sigma_{D,f}^f} \right)^{\frac{1}{\sigma_{D,f}^f - 1}}, \quad q_{\text{dep},k}^i r_{\text{dep},k} = q_{\text{dep}}^i r_{\text{dep}} \left( \frac{r_{\text{dep},k}}{r_{\text{dep}}} \right)^{\sigma_{D,f}^f}. \quad (30)$$

Under the optimal allocation,  $\mathcal{A}^i = R_A^i A^i$ , so  $R_A^i$  is the return per dollar of assets.

**Debt composition.** The firm finances its assets with equity  $E^i$  and a composite of imperfectly substitutable bank and nonbank debt. Let  $j \in J_l^F = \{b, nb\}$ . The effective funds borrowed are  $\mathcal{D}^i$ , a CES aggregate of debt classes:

$$\mathcal{D}^i = \left( \sum_{j \in J_l^F} \beta_j^i (q_j^i)^{\frac{\sigma_D^f - 1}{\sigma_D^f}} \right)^{\frac{\sigma_D^f}{\sigma_D^f - 1}}. \quad (31)$$

Analogous to the deposit class on the asset side, the bank-loan class is itself a CES aggregator over individual banks, with banks differentiated but symmetric:

$$q_b^i \equiv \left( \sum_{k \in K_f} (q_{b,k}^i)^{\frac{\sigma_{L,f}^f - 1}{\sigma_{L,f}^f}} \right)^{\frac{\sigma_{L,f}^f}{\sigma_{L,f}^f - 1}}, \quad (32)$$

where  $\sigma_{L,f}^f$  is the within-class elasticity of substitution across banks for firm loans. Optimally choosing borrowing quantities to minimize the cost of borrowing  $\sum_j q_j^i r_j$  yields

$$R_D^i = \left( \sum_{j \in J_l^F} (\beta_j^i)^{\sigma_D^f} r_j^{1-\sigma_D^f} \right)^{\frac{1}{1-\sigma_D^f}}, \quad q_j^i = \mathcal{D}^i (\beta_j^i)^{\sigma_D^f} \left( \frac{r_j}{R_D^i} \right)^{-\sigma_D^f}. \quad (33)$$

For the bank-loan class ( $j = b$ ), the class-level rate  $r_b$  entering the across-class debt demand above is itself a CES aggregate of individual bank loan rates, and class-level quantity allocates across banks according to within-class CES demand:

$$r_b = \left( \sum_{k \in K_f} r_{b,k}^{1-\sigma_{L,f}^f} \right)^{\frac{1}{1-\sigma_{L,f}^f}}, \quad q_{b,k}^i = q_b^i \left( \frac{r_{b,k}}{r_b} \right)^{-\sigma_{L,f}^f}. \quad (34)$$

Effective funding quantity and the cost of debt are related by  $\sum_{j \in J_l^F} r_j q_j^i = R_D^i \mathcal{D}^i$ . Under the assumption that firms face national-average bank rates within each class—that is,  $r_{\text{dep},k} = r_{\text{dep}}$  and  $r_{b,k} = r_b$  for all  $k$  in the firm-level data—the within-class elasticities  $\sigma_{D,f}^f$  and  $\sigma_{L,f}^f$  do not enter the across-class firm demand system. They are model primitives identified separately from bank rate passthrough rather than from firm-level data.

**Aggregation.** To pin down firm scale  $A^i$ , we assume that capital is freely mobile across firms and industries. Free mobility implies that investors can reallocate across firms until no firm's market value exceeds its replacement cost: if  $q > 1$  for some firm, capital flows in, expanding its scale and driving  $q$  back toward one. In aggregate, this is the standard free-entry condition applied to the real investment margin, and it requires that the economy-wide weighted average of Tobin's  $q$  equals one.

The market value of firm  $i$  is the discounted value of its accounting profits. Firm value and Tobin's average  $q$  are

$$V^i = \frac{\Pi^i}{r_E}, \quad q^i = \frac{V^i}{A^i}. \quad (35)$$

Imposing the equilibrium condition and letting  $w^i \equiv A^i / \sum_j A^j$  denote firm  $i$ 's economy-wide asset share gives

$$\sum_i w^i q^i = 1. \quad (36)$$

Simplifying and substituting in the optimal equity ratio yields the restriction

$$\sum_i w^i \left[ R_A^i - \left( r_E + R_D^i(1 - \bar{s}^i) - \frac{(R_D^i)^2}{2\eta_E^i} \right) \right] = 0, \quad (37)$$

which implicitly traces out the required return on assets for a given return on equity, coming from, for example, household demand to own equities.

We close the firm sector by assuming decreasing returns to real projects in aggregate. Letting  $A_{\text{real}}^s$  denote aggregate sector-level firm demand for real assets, real asset returns in sector  $s$  are

$$r_{\text{real}}^s = \bar{r}_{\text{real}} \left( \frac{A_{\text{real}}^s}{\bar{A}^s} \right)^{-\eta_K}. \quad (38)$$

For example, suppose there is a decrease in the required return to hold equity,  $r_E$ . For equation (37) to hold,  $R_A^i$  must also decrease. From equation (38), for real asset returns to fall, the quantity of real assets must increase. Thus, lower required equity returns lead to greater firm investment scale.

For the deposit and bank-loan classes, total firm demand at the bank level is obtained

by aggregating across firms:

$$Q_{\text{dep},k}^F = \sum_i q_{\text{dep},k}^i, \quad Q_{b,k}^F = \sum_i q_{b,k}^i. \quad (39)$$

### 4.3 Estimation

The estimation follows the nested structure of the firm problem and the empirical patterns documented in Section 4.1. The facts show that firm size and industry are associated with the allocation between real and financial assets, the composition of financial assets, leverage, and the allocation of debt across bank and nonbank credit. The model interprets these patterns as variation in firm demand across asset and debt classes. As in the household section, we separate elasticities from heterogeneity. Aggregate time-series variation in relative returns identifies the across-class elasticities, while QFR microdata identify how the implied preference weights vary across industry-by-size firm types. In the empirical implementation, the model index  $i$  corresponds to a firm type defined by industry and size.

Estimation proceeds in four stages.

**Stage 1: across-class elasticities from Financial Accounts time series.** The financial-asset demand equation (29) implies that, for any two financial-asset classes  $k$  and  $l$ ,

$$\log\left(\frac{F_{k,t}}{F_{l,t}}\right) = \sigma_F^f \log\left(\frac{\bar{\gamma}_k}{\bar{\gamma}_l}\right) + (\sigma_F^f - 1) \log\left(\frac{r_{k,t}}{r_{l,t}}\right) + \tilde{\varepsilon}_{k,l,t}, \quad (40)$$

where  $F_{k,t}$  is aggregate firm-sector holdings of financial-asset class  $k$  at quarter  $t$ ,  $r_{k,t}$  is the population-average return on class  $k$ , and  $\bar{\gamma}_k$  is the population-average preference weight for class  $k$ .<sup>2</sup> We estimate equation (40) as a two-class IV regression comparing MMF holdings to deposits using quarterly Financial Accounts data. Identification of  $\sigma_F^f$  comes from time-series variation in the relative return between MMFs and deposits, instrumented with Federal Reserve Treasury holdings and Federal Reserve Agency MBS holdings. The specification includes a post-2000 dummy to account for a structural break in the composition of firm liquidity.

The borrowing-side analog follows from equation (33). For bank and nonbank debt classes

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<sup>2</sup>At the aggregate level, the within-class price index for deposits collapses to the average deposit return when within-class preference weights are not parameterized as functions of firm characteristics. The within-class elasticities therefore do not enter equation (40).

$k$  and  $l$ ,

$$\log\left(\frac{\mathcal{D}_{k,t}}{\mathcal{D}_{l,t}}\right) = \sigma_D^f \log\left(\frac{\bar{\beta}_k}{\bar{\beta}_l}\right) - \sigma_D^f \log\left(\frac{r_{k,t}^D}{r_{l,t}^D}\right) + \tilde{\varepsilon}_{k,l,t}, \quad (41)$$

where  $\mathcal{D}_{k,t}$  is aggregate firm borrowing in debt class  $k$  and  $r_{k,t}^D$  is the corresponding borrowing rate. We estimate this equation using quarterly Financial Accounts data on the bank-versus-nonbank composition of firm debt. Identification of  $\sigma_D^f$  comes from time-series variation in the relative borrowing rate between bank and nonbank debt, instrumented with Federal Reserve Treasury holdings and Federal Reserve Agency MBS holdings. The specification includes a post-2000 dummy to account for a structural break in firm debt composition.

Table 5 presents the results.

**Table 5: Firm CES Estimation: Across-class Elasticities.** This table reports the Stage 1 across-class elasticity IV regressions for the financial-asset nest, deposits versus money market funds in column (1), and the debt nest, bank versus nonbank debt in column (2). The regressions are estimated using quarterly Financial Accounts time-series data, with  $\log(\text{Fed Treasury holdings})$  and  $\log(1+\text{Fed Agency MBS holdings})$  as instruments and a post-2000 dummy as an exogenous control. Column (1) reports the coefficient  $\sigma_F^f - 1$ ; column (2) reports the coefficient  $-\sigma_D^f$ . Standard errors are in parentheses.

|                          | Saving<br>$\sigma_F^f - 1$<br>(1) | Borrowing<br>$-\sigma_D^f$<br>(2) |
|--------------------------|-----------------------------------|-----------------------------------|
| $\log P_j/P_0$ (IV)      | 0.651***<br>(0.185)               | -1.482***<br>(0.323)              |
| First-stage F            | 63.2                              | 14.8                              |
| Observations             | 157                               | 173                               |
| Recovered $\hat{\sigma}$ | 1.651                             | 1.482                             |

**Stage 2: preference heterogeneity from QFR microdata.** Given  $\sigma_F^f$  from Stage 1, the financial-asset demand equation (29) can be rearranged at the industry-by-size firm-type level as

$$\log\left(\frac{F_{k,t}^i}{F_{l,t}^i}\right) - (\sigma_F^f - 1) \log\left(\frac{r_{k,t}}{r_{l,t}}\right) = \sigma_F^f \log\left(\frac{\gamma_k^i}{\gamma_l^i}\right) + \tilde{\varepsilon}_{k,l,t}^i. \quad (42)$$

We parameterize the relative financial-asset weights at the industry-by-size firm-type level:

$$\log\left(\frac{\gamma_k^i}{\gamma_l^i}\right) = \gamma_{k,l}^0 + \gamma_{k,l}^{\text{size}} \text{Size}^i + \gamma_{k,l}^{\text{ind}} \text{Industry}^i. \quad (43)$$

Given  $\sigma_F^f$  and aggregate class returns, equation (42) maps QFR asset shares into relative financial-asset weights. OLS estimates equation (43), recovering how firm size and industry shift demand between deposits and market-based financial assets. In the implementation, size enters as continuous mean-centered  $\log(\text{Size}_M)$  and its square, while industry enters through fixed effects. The recovered type-specific weights  $\gamma_{\text{dep}}^i$  and  $\gamma_{\text{mmf}}^i$  are stored for use in the quantitative model.

The borrowing side proceeds analogously. Given  $\sigma_D^f$  from Stage 1, equation (33) implies

$$\log\left(\frac{\mathcal{D}_{k,t}^i}{\mathcal{D}_{l,t}^i}\right) + \sigma_D^f \log\left(\frac{r_{k,t}^D}{r_{l,t}^D}\right) = \sigma_D^f \log\left(\frac{\beta_k^i}{\beta_l^i}\right) + \tilde{\varepsilon}_{k,l,t}^i. \quad (44)$$

We parameterize the relative borrowing weights as

$$\log\left(\frac{\beta_k^i}{\beta_l^i}\right) = \beta_{k,l}^0 + \beta_{k,l}^{\text{size}} \text{Size}^i + \beta_{k,l}^{\text{ind}} \text{Industry}^i. \quad (45)$$

OLS estimates equation (45), mapping the lender-choice patterns documented in Section 4.1 into size- and industry-specific demand for bank versus nonbank debt. In the implementation, size enters as continuous mean-centered  $\log(\text{Size}_M)$  and its square, while industry enters through fixed effects. The recovered type-specific weights  $\beta_{\text{bank}}^i$  and  $\beta_{\text{nonbank}}^i$  are stored for use in the quantitative model.

Table 6 presents the results.

**Table 6: Firm CES Estimation: Cell-level Preference Recovery.** This table reports cell-level preference-weight recovery. Given the estimated elasticities, the relative weights  $\gamma$  in column (1), deposits versus MMFs;  $\alpha$  in column (2), real versus financial assets; and  $\beta$  in column (3), bank versus nonbank debt, are inverted from the structural CES equations and regressed on continuous mean-centered  $\log(\text{Size})$  and  $\log(\text{Size})^2$  with industry fixed effects, weighted by total assets. Standard errors are in parentheses.

|                       | Saving                                          |                                                  | Borrowing                  |
|-----------------------|-------------------------------------------------|--------------------------------------------------|----------------------------|
|                       | $\log(\gamma_{\text{mmf}}/\gamma_{\text{dep}})$ | $\log(\alpha_{\text{real}}/\alpha_{\text{fin}})$ | $\log(\beta_b/\beta_{nb})$ |
|                       | (1)                                             | (2)                                              | (3)                        |
| $\log(\text{Size})$   | 0.153***<br>(0.016)                             | 0.228***<br>(0.028)                              | -0.297***<br>(0.017)       |
| $\log(\text{Size})^2$ | 0.011<br>(0.010)                                | 0.050***<br>(0.017)                              | -0.130***<br>(0.010)       |
| Industry FE           | Yes                                             | Yes                                              | Yes                        |
| Observations          | 830                                             | 750                                              | 830                        |
| R <sup>2</sup>        | 0.550                                           | 0.791                                            | 0.951                      |

**Stage 3: real versus financial assets.** After estimating the financial-asset weights, we construct the firm-specific financial return index  $R_{\text{fin}}^i$  from equation (29). The outer asset demand equation (28) implies

$$\log\left(\frac{q_{\text{real},t}^i}{F_t^i}\right) = \sigma_A^f \log\left(\frac{\alpha_{\text{real}}^i}{\alpha_{\text{fin}}^i}\right) + (\sigma_A^f - 1) \log\left(\frac{r_{\text{real},t}}{R_{\text{fin},t}^i}\right) + \varepsilon_t^i. \quad (46)$$

We parameterize the relative real-versus-financial asset weights at the industry-by-size firm-type level:

$$\log\left(\frac{\alpha_{\text{real}}^i}{\alpha_{\text{fin}}^i}\right) = \alpha^0 + \alpha^{\text{size}} \text{Size}^i + \alpha^{\text{ind}} \text{Industry}^i. \quad (47)$$

The coefficient on the relative return identifies  $\sigma_A^f$ , while variation by firm size and industry identifies heterogeneity in the relative demand for real versus financial assets. Because  $r_{\text{real},t}$  and  $R_{\text{fin},t}^i$  may respond to firm demand, we instrument the relative return with the Bauer–Swanson monetary-policy shock series and Federal Reserve Treasury holdings. The regression is estimated at the firm-type-by-year level, weighted by total assets, and includes continuous mean-centered  $\log(\text{Size}_M)$ , its square, and industry fixed effects.

Table 7 presents the results.

**Table 7: Firm CES Estimation: Outer Asset Nest.** This table reports the Stage 3 outer asset-nest IV regression for  $\sigma_A^f$ , estimated at the industry-by-size cell level. The dependent variable is the log ratio of real to financial assets, and the coefficient on the fitted relative return equals  $\sigma_A^f - 1$ . The relative return is instrumented using the Bauer–Swanson monetary-policy shock and Federal Reserve Treasury holdings. The regression includes industry fixed effects and continuous size controls  $\log(\text{Size})$  and  $\log(\text{Size})^2$ . Standard errors are in parentheses. \*p<0.1, \*\*p<0.05, \*\*\*p<0.01.

|                                                 | Log real/financial ratio<br>$\sigma_A^f - 1$ |
|-------------------------------------------------|----------------------------------------------|
| $\log(r_{\text{real}}/R_{\text{fin}})$ (fitted) | −0.254***<br>(0.026)                         |
| $\log(\text{Size})$                             | 0.170***<br>(0.021)                          |
| $\log(\text{Size})^2$                           | 0.037***<br>(0.013)                          |
| Industry FE                                     | Yes                                          |
| Observations                                    | 750                                          |

**Stage 4: leverage and decreasing returns.** The financing-scale margin is disciplined by the equity-ratio condition in equation (23). Let  $s_{E,t}^i \equiv E_t^i/A_t^i$  denote the equity-to-asset ratio. The model implies

$$s_{E,t}^i = \bar{s}^i + \frac{R_{D,t}^i}{\eta_E^i} + \tilde{\varepsilon}_t^i. \quad (48)$$

We estimate this equation at the firm-type-by-year level. The cost of debt  $R_{D,t}^i$  is instrumented with the Bauer–Swanson monetary-policy shock series and Federal Reserve Treasury holdings. The regression is weighted by total assets and includes continuous mean-centered  $\log(\text{Size}_M)$ , its square, and industry fixed effects. The slope on the fitted cost of debt identifies  $1/\eta_E^i$ , while the fixed effects recover the type-specific target equity ratios  $\bar{s}^i$ .

The decreasing-returns parameter in the real-asset supply schedule (38) is estimated from a first-differenced regression of log sectoral real returns on lagged changes in log sectoral real assets. Let  $K_t^s$  denote the empirical counterpart of sector- $s$  real assets,  $A_{\text{real},t}^s$ . The use of lagged  $\Delta \log K_{t-1}^s$  mitigates contemporaneous co-determination of returns and capital. The regression includes industry fixed effects and a linear time trend:

$$\Delta \log r_{\text{real},t}^s = -\eta_K \Delta \log K_{t-1}^s + \tau t + \mu_s + \tilde{\varepsilon}_t^s. \quad (49)$$

The coefficient on lagged  $\Delta \log K_{t-1}^s$  identifies  $-\eta_K$ . We then recover the sector-specific productivity shifter as

$$c_t^s = \log r_{\text{real},t}^s + \hat{\eta}_K \log K_t^s, \quad (50)$$

which serves as a sufficient statistic for counterfactual exercises.

Table 8 presents the results.

**Table 8: Firm CES Estimation: Leverage and Decreasing Returns.** This table reports Stage 4 estimates. Column (1) reports the leverage-equation IV regression used to recover  $\hat{\kappa}$ , where the dependent variable is the equity ratio and the coefficient on the fitted cost of debt equals  $1/\hat{\kappa}$ . The cost of debt is instrumented using the Bauer–Swanson monetary-policy shock and Federal Reserve Treasury holdings. Column (2) reports the first-differenced regression used to recover the decreasing-returns parameter  $\hat{\eta}$ , where the coefficient on lagged  $\Delta \log K$  equals  $-\hat{\eta}$ . Both regressions include industry fixed effects; column (1) also includes continuous size controls  $\log(\text{Size})$  and  $\log(\text{Size})^2$ . Standard errors are in parentheses. \*p<0.1, \*\*p<0.05, \*\*\*p<0.01.

|                       | $E/(E + D)$          | $\Delta r_{\text{real}}$ |
|-----------------------|----------------------|--------------------------|
|                       | (1)                  | (2)                      |
| $R_D$ (IV)            | 0.352***<br>(0.064)  |                          |
| $\log(\text{Size})$   | -0.004***<br>(0.001) |                          |
| $\log(\text{Size})^2$ | -0.002***<br>(0.001) |                          |
| $\Delta \log K_{t-1}$ |                      | -0.246***<br>(0.041)     |
| $t$                   |                      | 0.001**<br>(0.000)       |
| Industry FE           | Yes                  | Yes                      |
| Observations          | 750                  | 786                      |

**Identification, summarized.** The nested structure yields the same division of identification as on the household side. The across-class elasticities  $\sigma_F^f$  and  $\sigma_D^f$  are identified from time-series variation in relative returns in quarterly Financial Accounts data, using Federal Reserve balance-sheet quantities as instruments and a post-2000 dummy to absorb structural changes in firm balance-sheet composition. Given these elasticities, QFR microdata identify how firm size and industry shift relative demand across deposits versus MMFs, bank versus

nonbank debt, and real versus financial assets by inverting the corresponding CES demand equations and regressing the recovered weights on size controls and industry fixed effects.

The outer asset elasticity  $\sigma_A^f$  is identified from variation in the relative return between real and financial assets, instrumented by the Bauer–Swanson monetary-policy shock series and Federal Reserve Treasury holdings. The leverage parameters  $(\bar{s}^i, \eta_E^i)$  are identified from the relationship between equity ratios and debt costs, while the decreasing-returns parameter  $\eta_K$  is identified from the relationship between sectoral real returns and lagged changes in real assets. The within-class elasticities  $\sigma_{D,f}^f$  and  $\sigma_{L,f}^f$  do not enter this section’s estimation because QFR moments use aggregate class returns rather than firm-specific within-class CES price indexes; they are estimated with the intermediary-side parameters in Section 5.

## 5 Intermediaries

We now turn to the intermediary sector. The household and firm sections specify demand for savings and borrowing instruments. This section specifies the institutions that supply those instruments and determines the prices at which they clear. Banks issue deposits and make loans under monopolistic competition, subject to liquidity and capital wedges. Money market funds (MMFs) and non-bank lenders (NBLs) are competitive intermediaries that transform marketable securities into MMF shares and nonbank credit. Together, these intermediaries map household and firm demand into equilibrium returns, spreads, and balance-sheet quantities.

### 5.1 Intermediary setup

#### 5.1.1 Banks

We model a representative bank  $B$  that operates in a market with imperfect substitution across banks. Banks are differentiated but symmetric: from the perspective of any sector  $j \in \{h, f\}$ , total deposits placed at the banking sector are  $D^j = \sum_{B'} D_{B'}^j$ , with corresponding within-class CES price index  $R_{D,j}$  governed by within-deposit elasticity  $\sigma_{D,j}$  across banks. Loans to sector  $j$  are similarly aggregated with within-loan elasticity  $\sigma_{L,j}$ . The representative bank we model is one of many in this nested structure; aggregate bank balance-sheet quantities in our data correspond to the sum across banks.

Bank assets consist of bank loans made to households and firms,  $(L_B^h, L_B^f)$ , and a liquid asset composite  $S_B$ . These assets are financed by bank deposits issued to households and

firms,  $(D_B^h, D_B^f)$ , and an exogenous quantity of insider equity  $E_B$ . The bank's balance sheet equation is

$$\underbrace{L_B^h + L_B^f + S_B}_{\text{Bank assets}} = \underbrace{D_B^h + D_B^f + E_B}_{\text{Bank liabilities}}. \quad (51)$$

The liquid asset portfolio is a composite of imperfectly substitutable marketable assets: reserves ( $r$ ), Treasuries ( $g$ ), Agency MBS ( $h$ ), corporate bonds ( $c$ ), and NBL securities ( $n$ ),  $S_B^j$ ,  $j \in J_B \equiv \{r, g, h, c, n\}$ . Each asset has beginning-of-period price  $p^j$ , and the portfolio delivers  $\mathcal{S}_B$  units of end-of-period payoff according to the CES composite

$$\mathcal{S}_B(\{S_B^j\}_j) = \left( \sum_{j \in J_B} \gamma_B^j (S_B^j)^{\frac{\eta_B-1}{\eta_B}} \right)^{\frac{\eta_B}{\eta_B-1}}. \quad (52)$$

Cost minimization for a given end-of-period payoff yields the standard CES price index and individual asset demand,

$$P_B = \left( \sum_{j \in J_B} \gamma_B^j (p^j)^{1-\eta_B} \right)^{\frac{1}{1-\eta_B}}, \quad S_B^j = \gamma_B^j \left( \frac{p^j}{P_B} \right)^{-\eta_B} \frac{S_B}{P_B}. \quad (53)$$

The gross payoff per dollar of liquid assets is therefore  $1/P_B$ , with net payoff  $r_B^S \equiv 1/P_B - 1$ .

Liquidity and capital regulation are modeled as shadow costs of deviation from target ratios  $\lambda$  and  $\kappa$ :

$$\mathcal{L}(S_B, D_B^h, D_B^f) = \frac{\beta_L}{2} \left( \lambda - \underbrace{\frac{S_B}{D_B^h + D_B^f}}_{\equiv \psi} \right)^2, \quad \mathcal{K}(E_B, L_B^h, L_B^f) = \frac{\beta_K}{2} \left( \kappa - \underbrace{\frac{E_B}{L_B^h + L_B^f}}_{\equiv \phi} \right)^2. \quad (54)$$

The bank chooses product rates  $(r_B^{hd}, r_B^{fd}, r_B^{hl}, r_B^{fl})$  to maximize net cash flows to inside equity, subject to the resulting balance-sheet quantities. Demand for the bank's products from each sector follows the within-class CES system established in Sections 3 and 4. Holding the within-class deposit price index  $R_{D,j}$  and aggregate sectoral deposits  $D^j$  fixed, the bank's

deposit demand from sector  $j$  is

$$D_B^j = D^j \cdot \left( \frac{r_B^{jd}}{R_{D,j}} \right)^{\sigma_{D,j}-1}, \quad (55)$$

and analogously for loans. The bank's profit function is

$$\begin{aligned} \Pi_B = & (r_B^S - r_B^{hd} - c_B^{hd})D_B^h + (r_B^S - r_B^{fd} - c_B^{fd})D_B^f \\ & + (r_B^{hl} - r_B^S - c_B^{hl})L_B^h + (r_B^{fl} - r_B^S - c_B^{fl})L_B^f \\ & - \mathcal{L}(S_B, D_B^h, D_B^f) - \mathcal{K}(E_B, L_B^h, L_B^f) + r_B^S E_B, \end{aligned} \quad (56)$$

where  $(c_B^{hd}, c_B^{fd}, c_B^{hl}, c_B^{fl})$  are per-unit operating costs. The household-side bank-level demand quantities  $(D_B^h, L_B^h)$  correspond to the bank- $B$  deposit and loan positions from the household problem aggregated over households; the firm-side quantities are defined analogously.

We assume that each bank is small relative to its product market, so that it does not internalize the impact of its own pricing on the within-class CES price indices  $R_{D,j}$  and  $R_{L,j}$ .<sup>3</sup> Under this assumption, the bank's perceived demand elasticity for each product equals the corresponding within-class elasticity. The first-order conditions yield the optimal pricing relationships for sector  $j \in \{h, f\}$ :

$$r_B^{jd} = \left( \frac{\sigma_{D,j} - 1}{\sigma_{D,j}} \right) \left[ r_B^S + \beta_L(\lambda - \psi) \frac{1 - \psi}{D_B} - c_B^{jd} \right], \quad (57)$$

$$r_B^{jl} = \left( \frac{\sigma_{L,j}}{\sigma_{L,j} - 1} \right) \left[ r_B^S + \beta_L \frac{\lambda - \psi}{D_B} + \beta_K \frac{\phi}{L_B} (\kappa - \phi) + c_B^{jl} \right]. \quad (58)$$

Each  $\sigma_{D,j}$  or  $\sigma_{L,j}$  governs substitution *within* the corresponding product market: across banks for depositors of sector  $j$ , or across banks for borrowers of sector  $j$ . These are distinct from the across-class elasticities estimated in Sections 3 and 4, which govern substitution between deposits and MMFs, or between bank and non-bank borrowing.

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<sup>3</sup>This is the standard Dixit–Stiglitz monopolistic-competition assumption. Under this assumption, a bank changing  $r_B^{jd}$  moves only its own quantity  $D_B^j$ , holding  $R_{D,j}$  and  $D^j$  fixed; equivalently, the bank takes the deposit-market price index as given. Without this assumption the perceived elasticity becomes  $\sigma_{D,j} - \omega_B(\sigma_{D,j} - \sigma_S^j)$ , where  $\omega_B$  is the bank's market share within deposits and  $\sigma_S^j$  is the across-class savings elasticity. Standard practice in the deposit-channel literature (?) maintains the no-internalization assumption.

### 5.1.2 Money market funds

Money market funds own a composite of liquid securities, with beginning-of-period value  $S_M$ , which they finance entirely by issuing money market fund shares. They incur a per-unit operating cost  $c_M$ . The sector is competitive and individual money market funds earn zero profits.

The liquid asset portfolio is a composite of imperfectly substitutable safe, liquid assets: overnight repurchase agreements ( $o$ ), Agency MBS ( $h$ ), and Treasuries ( $g$ ), with quantities  $S_M^j$ ,  $j \in J_M \equiv \{o, h, g\}$ . Each asset has beginning-of-period price  $p^j$ . The portfolio delivers  $S_M$  units of end-of-period payoff according to the CES composite

$$\mathcal{S}_M(\{S_M^j\}_j) = \left( \sum_{j \in J_M} \gamma_M^j (S_M^j)^{\frac{\eta_M-1}{\eta_M}} \right)^{\frac{\eta_M}{\eta_M-1}}, \quad S_M = \sum_{j \in J_M} p^j S_M^j. \quad (59)$$

Cost minimization for a given end-of-period payoff yields the standard CES price index and demand,

$$P_M = \left( \sum_{j \in J_M} \gamma_M^j (p^j)^{1-\eta_M} \right)^{\frac{1}{1-\eta_M}}, \quad S_M^j = \gamma_M^j \left( \frac{p^j}{P_M} \right)^{-\eta_M} \frac{S_M}{P_M}. \quad (60)$$

The gross payoff from the optimal portfolio per dollar invested is therefore  $1/P_M$ , with net payoff  $r_M \equiv 1/P_M - 1$ . MMFs issue shares one-to-one with their assets, with promised return  $r_M^{ud}$ . MMF profit per unit of assets is

$$\Pi_M = \underbrace{(1 + r_M)}_{\text{portfolio payoff}} - \underbrace{(1 + r_M^{ud})}_{\text{investor payoff}} - \underbrace{c_M}_{\text{operating cost}}. \quad (61)$$

The zero-profit condition implies  $r_M^{ud} = r_M - c_M$ . Aggregate demand for MMF shares pins down the scale of the MMF sector:

$$S_M = D_M^{ud} \equiv \sum_i q_i^{hm} + \sum_i q_i^{fm}, \quad (62)$$

where  $q_i^{hm}$  and  $q_i^{fm}$  are the MMF holdings from the household and firm problems. The MMF class is treated as a single market claim, so it has no within-class structure across funds.

### 5.1.3 Non-bank lenders

Non-bank lenders make loans and finance them with a composite of bank-held NBL securities and broadly held security issuance. There are two types of NBLs: a household lender, indexed by  $h$ , and a firm lender, indexed by  $f$ . The two types specialize in which sector they lend to but are otherwise symmetric. The sector is competitive and individual lenders earn zero profits.

Each type  $k \in \{h, f\}$  lends an amount  $L_{L,k}$  and receives an end-of-period payoff  $(1 + r_{L,k})L_{L,k}$ . The financing composite for household lenders consists of  $j \in J_h \equiv \{h, n\}$ , where  $h$  denotes broadly held securities financed by household loans, such as Agency MBS, and  $n$  denotes NBL securities held by banks. The financing composite for firm lenders consists of  $j \in J_f \equiv \{c, n\}$ , where  $c$  denotes broadly held securities financed by corporate loans, such as corporate bonds, and  $n$  again denotes NBL securities held by banks.<sup>4</sup> NBLs have a per-unit operating cost  $c_{L,k}$ .

Each NBL type issues  $S_{L,k}^j$  units of financing instruments, with required repayment  $p^j = 1 + r^j$  at the end of the period. Issuing  $\{S_{L,k}^j\}$  raises  $\mathcal{S}_{L,k}$  effective units of funding via the CES composite

$$\mathcal{S}_{L,k} = \left( \sum_{j \in J_k} \gamma_{L,k}^j (S_{L,k}^j)^{\frac{\eta_{L,k}-1}{\eta_{L,k}}} \right)^{\frac{\eta_{L,k}}{\eta_{L,k}-1}}. \quad (63)$$

Cost minimization for a given effective funding raised yields the standard CES price index and demand,

$$P_{L,k} = \left( \sum_{j \in J_k} \gamma_{L,k}^j (p^j)^{1-\eta_{L,k}} \right)^{\frac{1}{1-\eta_{L,k}}}, \quad S_{L,k}^j = \gamma_{L,k}^j \left( \frac{p^j}{P_{L,k}} \right)^{-\eta_{L,k}} \frac{S_{L,k}}{P_{L,k}}. \quad (64)$$

To raise one dollar of effective funding, the NBL is required to repay  $P_{L,k} \equiv 1 + \tilde{r}_{L,k}$  at the end of the period. NBL profit per unit of investment is

$$\Pi_{L,k} = (1 + r_{L,k}) - (1 + \tilde{r}_{L,k}) - c_{L,k}. \quad (65)$$

The zero-profit condition implies  $r_{L,k} = \tilde{r}_{L,k} + c_{L,k}$ . Given the funding rate, the scale of each

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<sup>4</sup>We abstract from repo funding for NBLs. In the data, repo is primarily a short-term money-market instrument intermediated by MMFs and the central bank, affecting our model through the MMF portfolio rather than as a funding source for NBL loan books.

NBL sector is pinned down by aggregate borrowing demand:

$$\mathcal{S}_{L,h} = \sum_i q_i^{hl}, \quad \mathcal{S}_{L,f} = \sum_i q_i^{fl}, \quad (66)$$

where  $q_i^{hl}$  is household  $i$ 's NBL borrowing and  $q_i^{fl}$  is firm  $i$ 's NBL borrowing from the household and firm problems.

## 5.2 Estimation

The setup above leaves three sets of intermediary-side parameters to discipline. First, CES portfolio elasticities determine how banks, MMFs, and NBLs substitute across marketable assets and funding instruments. Second, the bank pricing equations require within-class elasticities for household and firm deposits and loans; these elasticities govern how strongly bank quantities respond to bank-specific rates and therefore determine bank markups. Third, zero-profit conditions for MMFs and NBLs, together with bank first-order conditions, discipline operating costs and regulatory wedges. We estimate these objects in turn.

### 5.2.1 Bank liquid asset portfolio

The bank holds a CES portfolio of liquid securities with common elasticity  $\sigma_B = \eta_B$ . Equation (53) implies

$$\log\left(\frac{S_j^B}{S_{\text{res}}^B}\right) = \log\left(\frac{a_j}{a_{\text{res}}}\right) + (1 - \sigma_B) \log\left(\frac{p_j}{p_{\text{res}}}\right), \quad (67)$$

where  $p_j = 1/(1 + r_j)$  is the price of security  $j$  and reserves serve as the numeraire. We estimate this equation by IV, instrumenting the endogenous relative price with Federal Reserve Treasury holdings and the federal funds rate interacted with asset-type indicators.

**Bank pricing and within-class elasticities.** Equations (57) and (58) imply passthrough relationships between the bank's shadow cost of funds, determined by the liquid-asset CES price index, and product-level rates. For loans,

$$r_L^x = \frac{\sigma_{L,x}}{\sigma_{L,x} - 1} (c_L^x + \tilde{r} + \text{wedge}), \quad x \in \{h, f\}, \quad (68)$$

where the wedge reflects liquidity and capital regulation costs. We estimate the four within-class passthrough elasticities  $\sigma_{L,h}$ ,  $\sigma_{L,f}$ ,  $\sigma_{D,h}$ , and  $\sigma_{D,f}$  using IV regressions in first differences. Specifically, we regress changes in product rates on changes in the bank shadow cost of funds, instrumenting with the federal funds rate for loan rates and monetary policy surprises for deposit rates. We then jointly estimate the regulatory parameters  $(\beta_L, \beta_K, \lambda, \kappa)$  and time-invariant operating cost margins using GMM, with orthogonality conditions between rate residuals and policy instruments in both levels and differences. Year-specific operating costs are inverted from the bank first-order conditions given the structural parameters.

These passthrough estimates identify the within-class elasticities under the structural interpretation of the bank model: each captures how strongly a marginal bank's deposit or loan quantity responds to its own rate, holding fixed the CES price index for the relevant within-class market.

The bank results are shown in Table 9. Panel A reports the securities portfolio CES estimation; Panel B reports the rate passthrough IV regressions. Figure ?? plots the time series of bank operating costs by product.

**Table 9: Bank Estimation.** Panel A estimates the bank's CES demand for securities. The dependent variable is the log portfolio share of security  $j$  relative to the composite ( $\log S_j^B/S_B$ ). Column (1) is OLS; column (2) instruments for the relative price with a monetary policy shock. Panel B estimates the passthrough of the bank's shadow cost of funds ( $\tilde{r}$ ) to product-level rates via IV using the federal funds rate (loans) or monetary policy surprises (deposits) as instruments. The dependent variable in each column is the change in the indicated rate ( $\Delta r$ ). \*p<0.1; \*\*p<0.05; \*\*\*p<0.01.

**Panel A: Securities Portfolio Demand**

|                          | $\log(S_j^B/S_B)$     |                       |
|--------------------------|-----------------------|-----------------------|
|                          | OLS                   | IV                    |
|                          | (1)                   | (2)                   |
| $\log(p_j/P_B)$          | -17.154***<br>(2.217) |                       |
| $\log(p_j/P_B)$ (fitted) |                       | -21.774***<br>(2.734) |
| Treasury                 | -1.297***<br>(0.080)  | -1.378***<br>(0.085)  |
| Agency                   | -0.422***<br>(0.090)  | -0.540***<br>(0.099)  |
| Corporate                | -1.471***<br>(0.095)  | -1.603***<br>(0.105)  |
| Nonbank                  | -2.892***<br>(0.097)  | -3.031***<br>(0.108)  |
| Pre-crisis               | 2.984***<br>(0.063)   | 3.019***<br>(0.064)   |
| Observations             | 756                   | 756                   |
| R <sup>2</sup>           | 0.810                 | 0.809                 |

**Panel B: Rate Passthrough (IV)**

|                             | Dep. var.: $\Delta r$ |                     |                  |                     |
|-----------------------------|-----------------------|---------------------|------------------|---------------------|
|                             | $\Delta r_{L,h}$      | $\Delta r_{L,f}$    | $\Delta r_{D,h}$ | $\Delta r_{D,f}$    |
|                             | HH Loans              | Firm Loans          | HH Deposits      | Firm Deposits       |
|                             | (1)                   | (2)                 | (3)              | (4)                 |
| $\Delta \tilde{r}$ (fitted) | 1.636***<br>(0.119)   | 1.602***<br>(0.116) | 0.267<br>(0.219) | 0.781***<br>(0.205) |
| Observations                | 35                    | 35                  | 35               | 35                  |
| R <sup>2</sup>              | 0.828                 | 0.829               | 0.475            | 0.803               |

**Money market funds.** We estimate the MMF CES portfolio system using equation (60). With repo as the numeraire, the demand system implies

$$\log\left(\frac{S_j^M}{S_{\text{repo}}^M}\right) = \log\left(\frac{\gamma_j^M}{\gamma_{\text{repo}}^M}\right) + (1 - \eta_M) \log\left(\frac{p_j}{p_{\text{repo}}}\right). \quad (69)$$

We instrument the endogenous relative price with Federal Reserve Treasury holdings interacted with the federal funds rate. Asset-type fixed effects recover the preference weights, and the IV coefficient identifies  $\eta_M$ . Operating costs are recovered from the zero-profit condition year by year:

$$c_{M,t} = r_{M,t} - r_{M,t}^{ud}. \quad (70)$$

**Non-bank lenders.** For the household NBL, the funding sources are agency securities and bank-held NBL securities. Equation (64) implies

$$\log\left(\frac{S_{\text{sec}}^{L,h}}{S_{\text{nbl}}^{L,h}}\right) = \log\left(\frac{\gamma_{\text{sec}}^{L,h}}{\gamma_{\text{nbl}}^{L,h}}\right) + (1 - \eta_L^h) \log\left(\frac{\bar{p}_{\text{age}}}{\bar{p}_{\text{nbl}}}\right), \quad (71)$$

where  $\bar{p}_j = 1 + r_j$  is the issuer price. For the firm NBL, the funding sources are corporate bonds and bank-held NBL securities, estimated analogously. Both regressions use IV with Federal Reserve Treasury holdings interacted with the federal funds rate as instruments. The coefficients identify  $\eta_L^h$  and  $\eta_L^f$ . Operating costs are recovered year by year from the zero-profit condition:

$$c_{L,k,t} = r_{L,k,t} - \tilde{r}_{L,k,t}. \quad (72)$$

The non-bank intermediary results are shown in Table 10, and Figure ?? plots the recovered operating costs over time.

**Table 10: Nonbank Intermediary Estimation.** Each column estimates the CES demand elasticity for a nonbank intermediary via IV. The dependent variable is the log portfolio share of asset  $j$  relative to the composite ( $\log S_j/S$ ). Column (1) estimates MMF demand across securities (Treasuries, agencies, other); column (2) estimates household nonbank lender (HH NBL) demand across funding sources (bank loans, agencies); column (3) estimates firm nonbank lender (Firm NBL) demand across funding sources (bank loans, bonds). \* $p < 0.1$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .

|                        | $\log(S_j/S)$              |                            |                          |
|------------------------|----------------------------|----------------------------|--------------------------|
|                        | MMF                        | HH NBL                     | Firm NBL                 |
|                        | (1)                        | (2)                        | (3)                      |
| $\log(p_j/P)$ (fitted) | $-13.637^{***}$<br>(4.472) | $-15.164^{***}$<br>(3.536) | $-4.878^*$<br>(2.675)    |
| Treasury               | $-2.126^{***}$<br>(0.087)  |                            |                          |
| Agency                 | $-2.194^{***}$<br>(0.115)  | $4.718^{***}$<br>(0.037)   |                          |
| Bond                   |                            |                            | $5.027^{***}$<br>(0.030) |
| First-stage F          | 751.6                      | 9997.6                     | 13833                    |
| Observations           | 322                        | 161                        | 161                      |
| $R^2$                  | 0.027                      | 0.146                      | 0.033                    |

### 5.3 Summary of intermediary parameter estimates

Table 11 consolidates the CES elasticities, preference weights, and calibrated parameters across all intermediary sectors.

**Table 11: Intermediary Parameter Estimates.** CES elasticities, preference weights, and calibrated parameters for banks, MMFs, and NBLs. Standard errors from IV regressions are reported where applicable. Operating costs are sample means in basis points.

| Parameter                                 | Description                        | Estimate | SE    |
|-------------------------------------------|------------------------------------|----------|-------|
| <b>Banks — Securities Portfolio</b>       |                                    |          |       |
| $\sigma_B$                                | Securities CES elasticity          | 22.774   | 2.734 |
| $a_{\text{res}}$                          | Reserve weight                     | 0.480    | —     |
| $a_{\text{tre}}$                          | Treasury weight                    | 0.121    | —     |
| $a_{\text{age}}$                          | Agency weight                      | 0.280    | —     |
| $a_{\text{cor}}$                          | Corporate bond weight              | 0.097    | —     |
| $a_{\text{nbl}}$                          | NBL securities weight              | 0.023    | —     |
| <b>Banks — Within-class passthrough</b>   |                                    |          |       |
| $\sigma_{Lh}$                             | HH loan passthrough                | 2.593    | 0.294 |
| $\sigma_{Lf}$                             | Firm loan passthrough              | 2.679    | 0.321 |
| $\sigma_{Dh}$                             | HH deposit passthrough             | 1.847    | 0.406 |
| $\sigma_{Df}$                             | Firm deposit passthrough           | 4.582    | 4.281 |
| <b>Banks — Regulation</b>                 |                                    |          |       |
| $\lambda$                                 | Liquidity ratio target             | 0.542    | —     |
| $\kappa_B$                                | Capital ratio target               | 0.172    | —     |
| $\beta_L$                                 | Liquidity cost parameter           | 0.009    | —     |
| $\beta_K$                                 | Capital cost parameter             | 0.011    | —     |
| <b>Banks — Mean Operating Costs (bps)</b> |                                    |          |       |
| $\bar{c}_{B,hl}$                          | HH loans                           | −97.4    | —     |
| $\bar{c}_{B,fl}$                          | Firm loans                         | −90.1    | —     |
| $\bar{c}_{B,hd}$                          | HH deposits                        | −93.9    | —     |
| $\bar{c}_{B,fd}$                          | Firm deposits                      | 89.6     | —     |
| <b>Nonbanks</b>                           |                                    |          |       |
| $\eta_{\text{MMF}}$                       | MMF CES elasticity                 | 14.637   | 4.472 |
| $\eta_{\text{NBL},h}$                     | HH nonbank lender elasticity       | 16.164   | 3.536 |
| $\eta_{\text{NBL},f}$                     | Firm nonbank lender elasticity     | 5.878    | 2.675 |
| $\bar{c}_M$                               | Mean MMF operating cost (bps)      | 53.4     | —     |
| $\bar{c}_{L,h}$                           | Mean HH NBL operating cost (bps)   | 92.5     | —     |
| $\bar{c}_{L,f}$                           | Mean firm NBL operating cost (bps) | 0.0      | —     |

## 6 Other Sectors and Equilibrium

We close the model with two sectors that primarily discipline aggregate asset supplies and external demand: the government and the rest of the world. We then state the market-clearing conditions that determine equilibrium returns and balance-sheet quantities.

### 6.1 Government

The government represents the consolidated balance sheet of the fiscal and monetary authority. It holds exogenous net positions in Treasuries and agency MBS, and it supplies reserves to banks and repo to MMFs perfectly elastically at policy rates set by the monetary authority. We do not impose a government budget constraint in the model. Net Treasury issuance, agency holdings, reserve supply, and repo supply are taken as exogenous, with policy rates serving as the relevant government-sector prices in counterfactual exercises.

### 6.2 Rest of the world

The rest of the world (RoW) consists of non-domestic investors that demand U.S. marketable claims. Let  $j \in J^W \equiv \{g, h, c, e\}$  index Treasuries, agency MBS, corporate bonds, and equity. Foreign demand follows a CES system:

$$P^W = \left( \sum_{j \in J^W} \alpha_j^W p_j^{1-\sigma^W} \right)^{\frac{1}{1-\sigma^W}}, \quad q_j^W = M^W \alpha_j^W \left( \frac{p_j}{P^W} \right)^{1-\sigma^W}, \quad (73)$$

where  $P^W$  is the RoW price index,  $\alpha_j^W$  is the preference weight on asset  $j$ ,  $p_j = 1/(1+r_j)$  is the price of asset  $j$ ,  $\sigma^W$  is the RoW elasticity of substitution,  $M^W$  is exogenous foreign financial wealth, and  $q_j^W$  is foreign demand for asset  $j$ . We do not model foreign borrowing, which accounts for a small share of U.S. financial assets.

**Estimation.** With Treasuries as the numeraire, the demand system implies

$$\log \left( \frac{S_j^{\text{ROW}}}{S_{\text{tre}}^{\text{ROW}}} \right) = \log \left( \frac{a_j^{\text{ROW}}}{a_{\text{tre}}^{\text{ROW}}} \right) + (1 - \sigma_{\text{ROW}}) \log \left( \frac{p_j}{p_{\text{tre}}} \right). \quad (74)$$

We estimate  $\sigma_{\text{ROW}}$  by IV, instrumenting the endogenous relative price with Federal Reserve Treasury holdings and the federal funds rate interacted with asset-type indicators. Asset-

type fixed effects recover relative preference weights. We recover the scale of foreign demand  $M^W$  and the absolute preference weights by matching observed 2019 quantities. Estimation results are reported in Table 12; the RoW elasticity estimate  $\hat{\sigma}^W \approx 18$  implies substantial substitutability across the marketable claims held by foreign investors.

### 6.3 Equilibrium

An equilibrium is a vector of returns and quantities such that households, firms, intermediaries, the government, and the rest of the world satisfy their optimality conditions and all asset markets clear. Household saving and borrowing choices are governed by the consumption-saving condition (11), the saving demand system (6), the borrowing demand system (7), and the cash-flow constraint (5). Firms satisfy the asset-side and debt-side demand systems (28)–(33), the equity-ratio condition (23), and the aggregation condition (37). Banks satisfy the balance sheet (51), liquid-asset demand (53), and pricing conditions (57)–(58). MMFs and NBLs satisfy their zero-profit conditions and CES portfolio or funding choices. Real asset returns satisfy (38), and foreign demand satisfies (73).

Markets clear as follows:

- *Deposits*: bank deposit supply equals household and firm deposit demand.
- *Bank loans*: bank loan supply equals household and firm bank-loan demand, including bank credit supplied to NBLs through NBL securities.
- *Nonbank loans*: NBL loan supply equals household and firm nonbank borrowing demand.
- *MMF shares*: MMF share supply equals household and firm demand for MMF shares.
- *Marketable debt claims*: supplies of Treasuries, agency MBS, corporate bonds, and NBL securities equal demand from households, banks, MMFs, NBLs, the RoW, and the government, as applicable.
- *Reserves and repo*: government-supplied reserves and repo, available perfectly elastically at exogenous policy rates, equal demand from banks and MMFs.
- *Firm equity*: firm equity supply equals equity demand from households and the RoW.
- *Real assets*: real asset demand by firms equals real asset supply, with returns determined by the decreasing-returns schedule.

**Table 12: Rest-of-World CES Demand Estimation.** The dependent variable is the log portfolio share of asset  $j$  relative to Treasuries ( $\log S_j^{\text{ROW}}/S_{\text{tre}}^{\text{ROW}}$ ). Column (1) is OLS; column (2) instruments for the relative price using futures and federal funds rate changes. Asset-type fixed effects (corporate bonds, equities) capture level differences in ROW demand across asset classes.

|                          | Dep. var.: $\log(S_j^{\text{ROW}}/S_{\text{tre}}^{\text{ROW}})$ |                       |
|--------------------------|-----------------------------------------------------------------|-----------------------|
|                          | OLS                                                             | IV                    |
|                          | (1)                                                             | (2)                   |
| $\log(p_j/P_d)$          | -9.269*<br>(4.691)                                              |                       |
| $\log(p_j/P_d)$ (fitted) |                                                                 | -17.083***<br>(5.539) |
| Corporate                | 1.073***<br>(0.067)                                             | 1.037***<br>(0.069)   |
| Equity                   | 1.151***<br>(0.194)                                             | 0.846***<br>(0.226)   |
| First-stage F            |                                                                 | 200.9                 |
| Observations             | 120                                                             | 120                   |
| R <sup>2</sup>           | 0.913                                                           | 0.910                 |

*Note:* \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

## 7 Counterfactual Exercises

We use the estimated market-equilibrium model to quantify how real-sector heterogeneity shapes financial intermediation. In all counterfactuals, intermediary technology, regulatory wedges, preference elasticities, policy variables, and institutional features are held fixed at their 2019 U.S. estimates. We vary only the distribution of household and firm types. The exercises therefore ask how financial structure changes when the real sector changes, holding the financial environment fixed.

The counterfactuals isolate two channels. Household heterogeneity affects banks primarily through the funding side, by shifting saving demand between deposits and market-based claims, and through household credit demand. Firm heterogeneity affects banks primarily through the asset side, by shifting borrowing demand between bank and nonbank credit. Together, these channels provide a demand-side foundation for the growth of alternative financing: a wealthier household sector demands more market-based claims, while changes in firm composition shift the relative demand for bank versus nonbank finance, including forms of private credit and private equity. We study these mechanisms through comparative statics around the 2019 U.S. economy, a 1983–2019 historical exercise, and counterfactual U.S. economies whose real-sector distributions are shifted toward those of other countries.

We track three outcomes: the bank credit share, the bank funding share, and the bank liquidity ratio  $\psi = S_B/D_B$ . The bank credit share measures banks' role in supplying private financing. The bank funding share measures banks' role in absorbing savings through deposits. The liquidity ratio measures how banks allocate deposit funding between securities and loans; a lower  $\psi$  indicates that a larger share of bank balance sheets is committed to credit intermediation.

### 7.1 How Real-Sector Characteristics Shift Financial Structure

We first trace how individual real-sector characteristics affect equilibrium financial structure around the 2019 U.S. baseline. We vary five characteristics one at a time—three on the household side (old-age dependency, mean wealth, and the top-1% wealth share) and two on the firm side (firm-size dispersion and the services value-added share)—holding all other characteristics fixed at their 2019 values. Figures 11–12 report the resulting changes in the bank credit share, the bank funding share, and the bank liquidity ratio  $\psi$ .

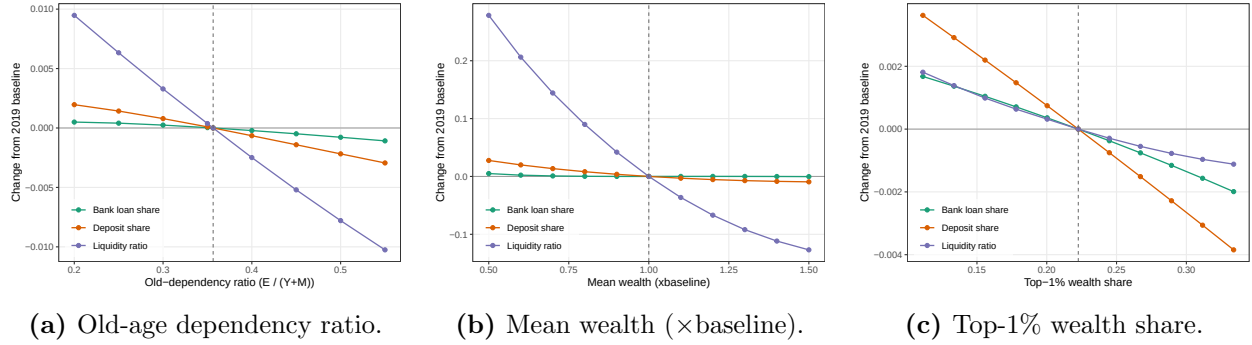
The household comparative statics show that household heterogeneity is central for banks' funding environment and balance-sheet composition. Mean wealth is the strongest force. In-

creasing mean household wealth reduces deposit reliance, because richer households allocate a larger share of savings directly to market-based claims. At the same time, higher wealth raises household borrowing capacity and credit demand. Banks respond by shifting assets away from securities and toward loans, so the liquidity ratio  $\psi$  falls sharply: scaling wealth from  $0.5\times$  to  $1.5\times$  the 2019 baseline moves  $\psi$  from 0.48 to 0.07. Wealth concentration and aging operate primarily through the portfolio-composition margin. A higher top-1% wealth share tilts aggregate saving away from deposits and toward direct securities, lowering the bank funding share; the old-age dependency ratio generates similar but smaller effects. Thus, household heterogeneity changes both sides of banks' balance sheets: it weakens the deposit franchise while increasing household credit demand.

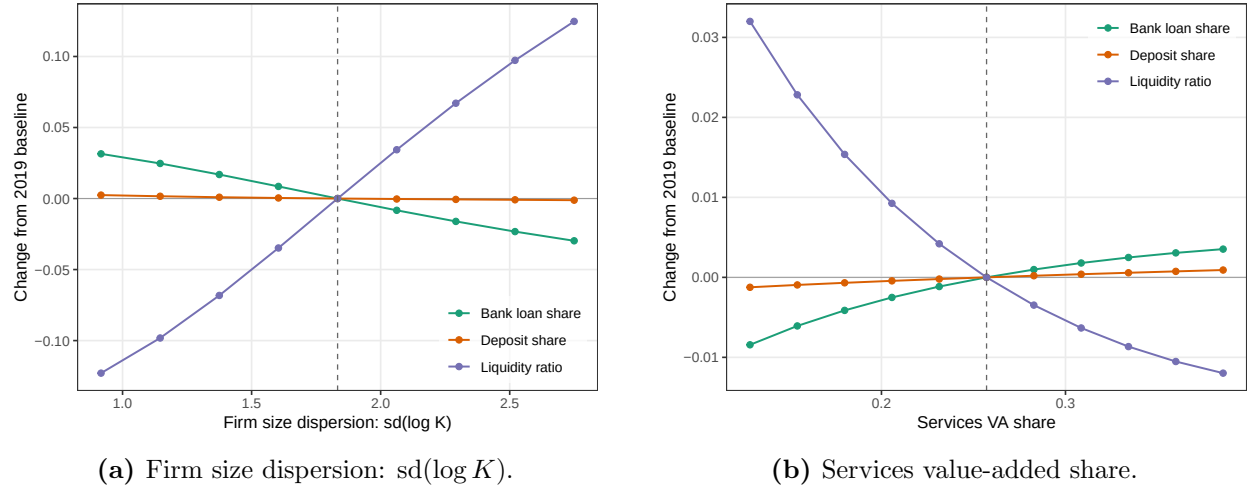
The firm comparative statics operate through a different margin. Firm-size dispersion mainly affects the asset side of banks' balance sheets. Increasing the dispersion of firm size shifts mass away from the mid-sized firms that rely most heavily on bank credit and toward small and large firms with lower bank-loan shares. This weakens bank credit demand and raises  $\psi$ , as banks hold more securities relative to deposits. The services share has relatively small effects on all three outcomes in the calibrated experiment, suggesting that the firm-side comparative static is driven more by the size distribution than by the coarse sectoral composition captured here.

Together, these comparative statics clarify the demand-side conditions under which alternative financing becomes more important. On the saver side, higher household wealth and greater concentration increase demand for market-based claims relative to bank deposits. On the borrower side, firm composition determines how strongly credit demand is tied to banks rather than to nonbank and market-based lenders. Alternative financing expands when both sides move in the same direction: savers demand claims outside the deposit system, and borrowers can be financed outside traditional bank lending relationships.

At the 2019 value of each characteristic, the model reproduces the calibrated 2019 equilibrium outcomes. Aggregate quantities of household financial assets and total private financing remain close to their baseline values across grid points, except in the mean-wealth sweep, where the size of the financial system scales with household wealth by construction. Because the reported outcomes are shares, the comparative statics isolate changes in financial structure rather than changes in aggregate scale.



**Figure 11: Comparative statics, household side.** Each panel varies one household characteristic around the 2019 baseline, holding all other characteristics fixed. The outcomes are the bank credit share, the bank funding share, and the bank liquidity ratio  $\psi$ , expressed as changes from the 2019 baseline.



**Figure 12: Comparative statics, firm side.** Each panel varies one firm characteristic around the 2019 baseline, holding all other characteristics fixed. The outcomes are the three measures in Figure 11.

## 7.2 Real-Sector Change and the Evolution of Financial Structure

We next ask how changes in the real sector contributed to the evolution of U.S. financial structure between 1983 and 2019. We solve four equilibria: the 2019 baseline, the 2019 economy with the 1983 household distribution, the 2019 economy with the 1983 firm distribution, and the 2019 economy with both household and firm distributions shifted to 1983.<sup>5</sup> In all cases, CES coefficients and intermediary parameters remain fixed at their 2019 estimates. Thus, the exercise isolates the effect of changes in the distribution of households and firms,

<sup>5</sup>1983 household cells are constructed by extrapolating the 1989 SCF backward using the SCF panel's wealth and income growth rates, mirroring the construction of `hh_cells.1980.extrapolated.csv`; switching to SCF+ historical microdata is a one-line change in the data-loading function. The implied scale factor is 0.28, meaning the 1983 household sector held 28% of the 2019 real wealth.

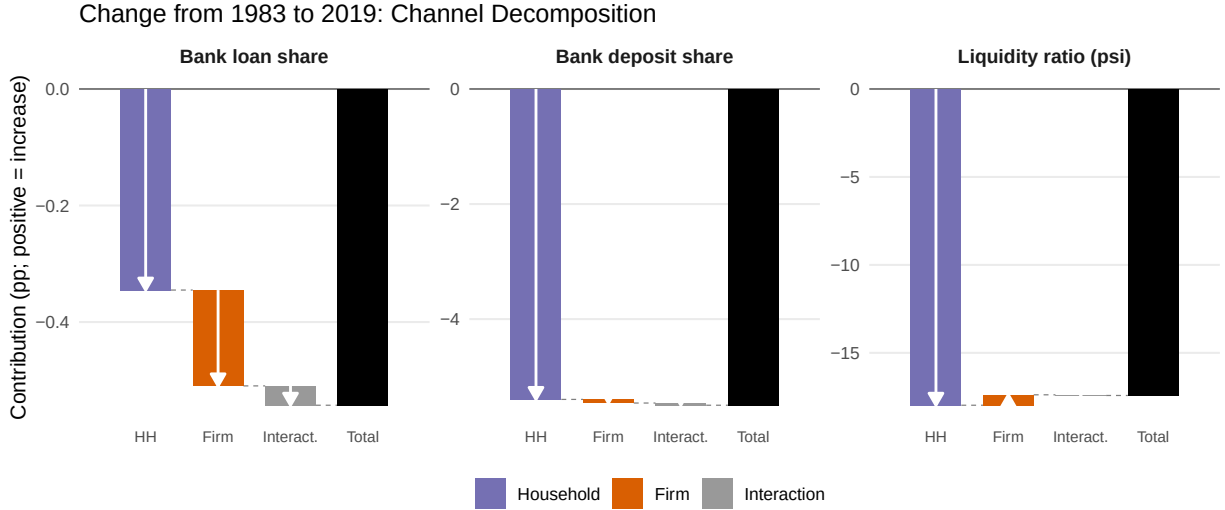
holding intermediary-side technology and institutions fixed.

The model implies that the long-run change in U.S. intermediation is primarily a household-side phenomenon. Moving from the 1983 real-sector distribution to the 2019 distribution leaves the bank credit share roughly unchanged ( $-0.5\text{pp}$ ), lowers the bank funding share by  $5.5\text{pp}$ , and reduces the bank liquidity ratio  $\psi$  by  $17.4\text{pp}$ . Replacing only the household distribution accounts for nearly all of these changes: the household-only counterfactual matches the joint shift to within  $0.6\text{pp}$  on every margin. By contrast, replacing only the firm distribution has little effect on the three aggregate outcomes.

The mechanism is the transformation of the household balance sheet. Rising real household wealth, together with changes in age structure and wealth concentration, increases demand for directly held market claims and reduces the share of saving absorbed by bank deposits. At the same time, higher household wealth expands borrowing capacity and credit demand, allowing banks to allocate more of their balance sheets to loans rather than securities. The result is a lower bank funding share and a lower  $\psi$ . The bank credit share changes little because both bank credit and total private financing scale with the size of the economy.

This result provides a demand-side interpretation of the broader rise of alternative financing. The model does not separately identify private credit or private equity as distinct sectors. But it does identify the real-sector conditions under which nonbank and market-based finance become more important. A wealthier household sector supplies more funding to market-based claims rather than deposits, while firm heterogeneity determines the extent to which borrowers can substitute away from traditional bank credit. These are precisely the demand-side forces that can support the growth of private credit, private equity, and other nonbank financing channels, alongside the supply-side innovations and institutional changes emphasized elsewhere in the literature.

This historical exercise reinforces the comparative statics in Section 7.1. Household wealth and its distribution dominate the effect of real-sector change on banks' funding role and balance-sheet composition, while firm heterogeneity matters primarily for the composition of credit demand. The major U.S. historical shift captured by the model therefore operates through the household side of banks' balance sheets, but its implications extend beyond banks: it changes the demand environment in which market-based and nonbank intermediaries compete with deposit-funded banks.



**Figure 13: 1983-to-2019: real-sector channels.** For each outcome, the bars show the contribution of household-side changes, firm-side changes, and their interaction to the model-implied change from 1983 to 2019. Positive values indicate that the outcome is higher in 2019 than in 1983. Household-side changes account for nearly the entire movement in the bank funding share and the bank liquidity ratio  $\psi$ ; firm-side and interaction effects are small.

### 7.3 Counterfactual Real-Sector Distributions from Other Countries

The first two exercises vary the U.S. real sector over local and historical changes. The final exercise uses cross-country variation in real-sector characteristics to construct a broader set of counterfactual U.S. economies. For each of the 20 countries with complete panel coverage, we replace the U.S. household and firm distributions with distributions that match key real-sector characteristics of that country, while keeping preference elasticities, intermediary-side parameters, policy variables, and institutional features fixed at their 2019 U.S. estimates. The exercise should therefore be interpreted as a real-sector counterfactual, not as a full model of each country’s financial system: it asks how U.S. financial structure would change if the U.S. real sector looked more like the real sector of another economy.

This distinction is important economically. Cross-country differences in financial structure reflect both demand-side forces and supply-side institutions. Demand-side forces arise because household wealth, demographics, firm size, and sectoral composition determine what kinds of financial claims households and firms want to hold or issue. Supply-side forces arise because banking regulation, creditor rights, bond-market depth, institutional investors, and financial infrastructure determine which intermediaries can efficiently supply and absorb

those claims. By holding the U.S. financial environment fixed, this exercise isolates the demand-side component: it asks how much cross-country variation in bank reliance can be generated by changing the real sector alone.

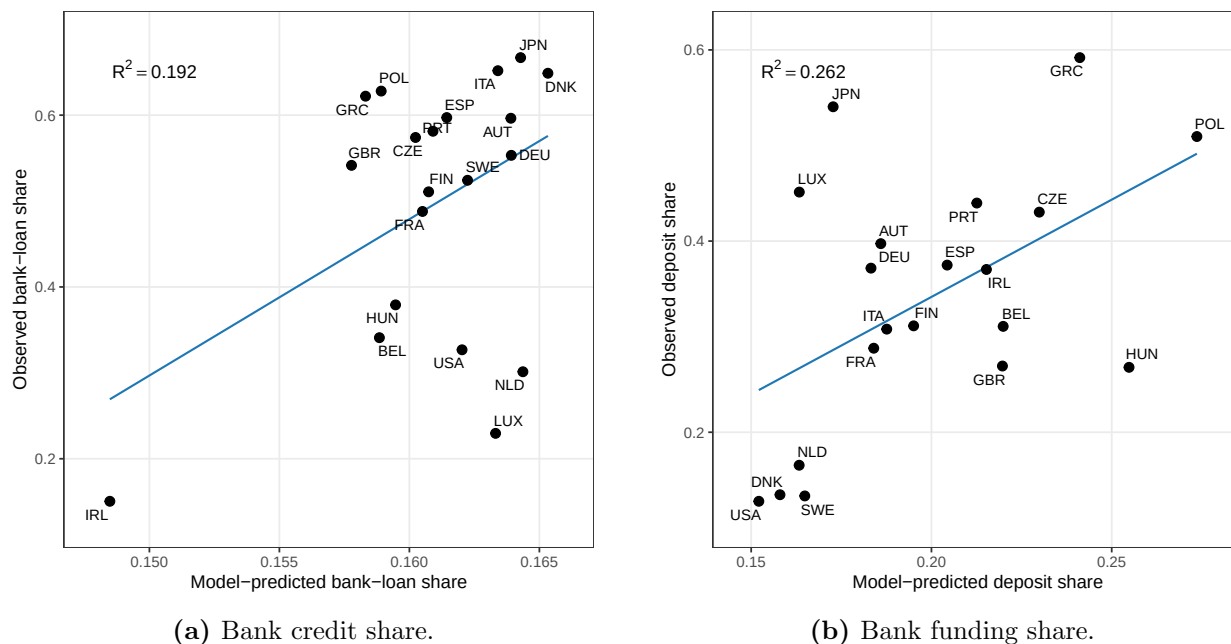
We construct each country’s inputs in two steps. On the household side, we start from the 2019 U.S. SCF cells, scale wealth and income by the country’s mean household wealth per capita, and reweight the population across age groups to match the country’s old-age dependency ratio. On the firm side, we shift productivity across services and non-services cells to match the country’s services value-added share, with an additional adjustment toward intangible-intensive sectors. Exogenous dollar quantities scale with the household wealth ratio. Because the country panel does not contain a comparable measure of banks’ securities-to-deposit ratio, we compare the model-implied bank credit share and bank funding share to their observed counterparts; model-implied  $\psi$  is reported only for reference.

Figure 14 shows that cross-country differences in real-sector characteristics generate meaningful variation in the model-implied financial structure. The model-implied shares are positively related to observed bank credit and funding shares, with  $R^2 = 0.19$  for the bank credit share and  $R^2 = 0.26$  for the bank funding share. The funding-side result has a direct economic interpretation. Richer and older household sectors hold a larger share of savings in market-based claims rather than deposits. When the U.S. model is given household distributions resembling those economies, bank funding shares fall. The model’s bivariate slope of the bank funding share on log household wealth is  $-0.04$ , compared with an observed slope of  $-0.10$ . Thus, real-sector differences alone generate a substantial share of the cross-country relationship between household development and deposit reliance, even before allowing intermediary institutions to vary.

The credit-side result is more limited and therefore more informative. The model generates the right directional variation in bank credit shares, but the magnitude is much smaller than in the data: the model-implied range is 1.7pp, compared with an observed range of 52pp. This gap indicates that firm and household demand characteristics alone do not determine how much credit flows through banks. In the U.S. institutional environment, firms with different size and industry composition shift the relative demand for bank versus non-bank debt, but the response is disciplined by U.S. elasticities, U.S. intermediary pricing, and U.S.-style access to market finance. Across actual countries, by contrast, the supply of bank and market credit is shaped by institutional differences that can greatly amplify or dampen the same real-sector demand forces. These differences include not only banking regulation and bond-market depth, but also the size and structure of nonbank institutional investors—

such as insurance companies, pension funds, mutual funds, and private funds—that absorb market-based claims and provide funding outside the deposit-funded banking system.

The economic interpretation is therefore asymmetric across margins. On the funding side, real-sector characteristics have substantial explanatory power because household portfolio demand directly determines the demand for deposits versus market-based saving instruments. On the credit side, real-sector characteristics matter, but their quantitative effect is constrained by the institutional environment that supplies credit. A country with large, intangible-intensive, or service-oriented firms may have high demand for market-based finance, but whether that demand translates into low bank reliance depends on the availability of bond markets, nonbank lenders, creditor protections, regulatory treatment of bank credit, and institutional investors such as insurers and pension funds that can hold and intermediate market-based claims. This is why the model can reproduce part of the cross-country pattern in bank credit shares while falling far short of the observed dispersion.



**Figure 14: Cross-country real-sector counterfactuals.** For each of the 20 countries with complete panel coverage, the figure plots observed shares against model-predicted shares from a counterfactual U.S. economy whose household and firm characteristics are shifted toward that country’s real-sector characteristics. Preference elasticities, intermediary-side parameters, policy variables, and institutional features remain fixed at their 2019 U.S. values. The solid line is OLS;  $R^2$  is annotated.

This exercise provides a disciplined way to use cross-country variation without interpreting the model as a full cross-country calibration. Real-sector differences, evaluated within

the U.S. financial environment, generate meaningful movement in bank funding and bank credit shares. But the two margins differ economically. Deposit reliance is closely tied to household portfolio demand, so changing the household distribution moves the model substantially toward the cross-country data. Bank credit reliance depends more heavily on the institutional supply of credit, so real-sector demand alone explains only a smaller structural component. The result reinforces the paper’s main message: real-sector characteristics are an important state variable for financial structure, but cross-country differences in banks, institutional investors, and market development remain essential, especially on the credit side.

## 8 Conclusion

This paper shows that the structure of financial intermediation is shaped not only by intermediary-side frictions, regulation, or financial development, but also by the characteristics of the real sector. Household net worth and age affect the demand for deposits, market-based saving instruments, and credit. Firm size and industry affect the demand for liquid assets, bank credit, and market-based debt. These demand forces determine the boundary between banks, markets, and nonbank intermediaries.

Using U.S. microdata, we estimate an equilibrium model in which heterogeneous households and firms choose among competing financial instruments supplied by banks, nonbank lenders, and markets. The counterfactuals show that real-sector heterogeneity is quantitatively important. Rising household wealth, aging, and wealth concentration reduce the role of bank deposits and reshape banks’ balance sheets, while firm heterogeneity primarily affects the composition of credit demand. Together, these forces provide a demand-side foundation for the growth of alternative financing, including forms of private credit and private equity.

At the same time, real-sector demand is not a substitute for institutional explanations. Cross-country real-sector counterfactuals explain a meaningful component of variation in bank funding and credit shares, but substantial residual dispersion remains, especially on the credit side. The boundary between banks and markets is therefore an equilibrium outcome shaped jointly by the characteristics of savers and borrowers and by the institutions that supply financial claims.